

# No-Regret Learning in Stackelberg Security Games with an Unknown, Bounded Set of Attacker Types

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## 1 Abstract

Stackelberg Security Games (SSGs) model the strategic allocation of limited defender resources against attackers whose preferences are often uncertain. Balcan et al. [2] designed no-regret online learning algorithms for the *repeated* variant of SSGs against an adversarial sequence of attackers drawn from a finite, *known* set  $K$ , and proved a matching linear-regret lower bound when  $K$  is unbounded. A natural real-world middle ground is left open: the defender does not know  $K$  a priori, but has a bound  $K_{\max}$  on its size. We first study this setting in the full-information feedback model. We propose GROW-CATALOGUE, an algorithm that incrementally learns new attacker types as they arrive, re-partitions the coverage polytope, and runs an anytime multiplicative-weights sub-routine on the resulting extreme-point set. Our central technical contribution is a full proof that GROW-CATALOGUE achieves expected regret

$$O\left(nK_{\max}\sqrt{T\log(nK_{\max})}\right)$$

against any oblivious adversarial sequence of length  $T$ . This is the rate of Balcan et al. [2] with  $k$  replaced by  $K_{\max}$ , inflated by the  $\sqrt{K_{\max}}$  that the unknown catalogue costs, and it is sublinear whenever  $n^2K_{\max}^2\log(nK_{\max}) = o(T)$ . Experiments on synthetic SSG instances confirm the predicted  $\sqrt{T}$  scaling, and the empirical regret stays comfortably below the analytic  $C\sqrt{T}$  envelope. We then ask whether the same relaxation is affordable when the defender observes only the attacked target, and show that it is not: under partial-information feedback with an unknown type set no algorithm achieves sublinear regret, even when one fixed type attacks in every round, because the defender can never learn *which* type it faces. What is missing is a name, so we identify the weakest supplement that supplies one, a periodic report naming the set of types seen so far, and prove that it restores regret  $O(nK_{\max}^{4/3}T^{2/3}\log^{1/3}(nK_{\max}) + K_{\max}B)$  for a report period  $B$ , the additive batching term being tight. Simulations of the lower-bound constructions match the theory to machine precision. Our implementation and all experiment scripts are at [github.com/aditisingh1053/Security\\_Games](https://github.com/aditisingh1053/Security_Games).

## 2 Introduction

In a Stackelberg Security Game (SSG), a defender allocates a scarce set of resources to protect  $n$  potential targets against a strategic attacker who, observing the defender’s randomized deployment, chooses the best target to attack [19]. Deployed systems built around this model include the US Coast Guard’s port protection patrols, the Federal Air Marshals assignment system, the LAX airport patrol schedule, and anti-poaching

patrols in several African national parks [19, 20]. In every one of these applications the defender must commit to its deployment before the attack happens, and the attacker’s payoff function encodes its preferences over targets.

The *classical* SSG analysis assumes that the attacker’s payoff function is fully known to the defender. The optimal commitment can then be computed by a linear program [e.g., 18]. In practice, however, the defender faces considerable uncertainty about the attacker. There may be several distinct threat actors (e.g., different poacher groups, different types of smugglers), and their utilities are only imperfectly known. A principled response, pioneered by Balcan et al. [2], is to drop the single-type assumption entirely and instead ask for a *no-regret* online algorithm against an adversarially chosen sequence of attacker types, drawn from a known finite set  $K$  with  $|K| = k$ . Under full-information feedback, they design an algorithm whose expected cumulative regret is bounded by  $O(\sqrt{Tn^2k\log(nk)})$ , which is sublinear in the horizon  $T$ , and they prove a matching linear-regret lower bound when  $|K|$  is unrestricted [2, Theorem 7.1].

The assumption that the set  $K$  itself is known to the defender in advance is, however, a strong one. In realistic deployments the defender typically does *not* have a complete list of attacker types at the start. What it may have is an *upper bound*  $K_{\max}$  on the number of distinct types it expects to encounter (for instance, from threat intelligence or from physical constraints of the domain). The question that motivates this project is therefore:

*Can the defender achieve sublinear regret when  $K$  is unknown a priori, subject to  $|K| \leq K_{\max}$ ?*

This question lies squarely in the gap left open between the *known- $K$*  upper bound and the *unrestricted- $K$*  lower bound of Balcan et al. [2]. It is not answered by any of the existing work, and resolving it is the main goal of the present report.

The answer turns out to depend sharply on what the defender is allowed to observe, and this report is organized around that dependence. Sections 3 and 4 set up the model and the unknown- $K$  problem, and Sections 5 and 6 give GROW-CATALOGUE together with its guarantee under full-information feedback. Section 7 then withdraws that feedback, leaving the defender only the attacked target, and shows that the question now has a negative answer: no algorithm is sublinear, even against a single fixed type. Section 8 identifies the weakest supplement that repairs this, a periodic report naming the types met so far, and recovers a sublinear rate, with matching lower bounds on what the batching costs. Section 9 reports the experiments and Section 10 the open problems.

### 2.1 Contributions

Our contributions are the following.

- **Problem formalization (Section 4).** We formalize the unknown- $K$  variant of the repeated Stackelberg security game, identifying the minimal modification of the model of Balcan et al. [2] that captures it.
- **Algorithm (Section 5).** We propose GROW-CATALOGUE, an online algorithm that at each round maintains a *catalogue*  $C_t \subseteq K$  of attacker types observed so far, recomputes the set  $\mathcal{E}(C_t; \epsilon)$  of  $\epsilon$ -approximate extreme points of the partition of  $\mathcal{P}$  induced by  $C_t$ , and runs an anytime multiplicative-weights sub-routine over  $\mathcal{E}(C_t)$ . When a new type is observed, the catalogue is updated and the sub-routine is restarted on the enlarged expert set.
- **Main theorem and a complete, detailed proof (Sections 5.2 and 6).** The technical core of this report is a self-contained proof that GROW-CATALOGUE achieves expected regret at most  $O(nK_{\max}\sqrt{T}\log(nK_{\max}))$  against any oblivious adversarial sequence of length  $T$  under full-information feedback. The rate matches Theorem 5.1 of Balcan et al. [2] with  $k$  replaced by  $K_{\max}$ ; for constant  $K_{\max}$  it is  $\tilde{O}(n\sqrt{T})$ .
- **Regime analysis (Corollaries 7 and 8).** We characterize the parameter regime in which the bound is meaningful ( $n^2K_{\max}^2\log(nK_{\max}) = o(T)$ ), and we prove an instance-adaptive refinement in which the leading factor depends on the number  $K'$  of types *actually* observed rather than on  $K_{\max}$ .
- **An impossibility result under partial information (Section 7).** We show that the two relaxations do not compose: if the defender observes only the attacked target and does not know  $K$ , then Theorem 12 gives expected regret at least  $T - 2$ , attained exactly, by Remark 13, and this holds already at  $K_{\max} = 1$ , so the obstruction is not the number of types but the defender's inability to *name* the one it faces. A separate phenomenon, Proposition 14, gives a  $\Theta(\sqrt{T})$  floor as soon as  $K_{\max} \geq 2$ , on a universe of only two types and under full information.
- **Block revelation (Section 8).** We formalize in Problem 15 the weakest supplement that repairs the impossibility, every  $B$  rounds an oracle reports the *set* of types seen so far, and nothing about when or how often they attacked, give Algorithm 2 for it, and prove in Theorem 18 a regret bound of  $O(nK_{\max}^{4/3}T^{2/3}\log^{1/3}(nK_{\max}) + K_{\max}B)$ . By Corollary 21 the report need arrive only  $\tilde{O}(T^{1/3})$  times over the horizon, and not fewer.
- **Matching lower bound for the relaxation (Theorem 22).** The additive  $\Theta(K_{\max}B)$  above cannot be removed: a nested threshold-interval construction inside  $G_3$  forces regret at least  $K_{\max}(B - 2)$  on every block-revelation algorithm, bracketing the batching cost as  $\Theta(K_{\max}B)$ .
- **Experiments (Section 9).** We implement GROW-CATALOGUE and two reference baselines and evaluate them on synthetic SSG instances; the empirical regret scales as predicted by the theorem, staying comfortably below the analytic  $C\sqrt{T}$  envelope. An oracle reformulation (Appendix B) removes the extreme-point enumeration and carries both algorithms to  $n = 10$  and  $n = 20$ , and we simulate each lower-bound construction exactly, confirming that its constant is attained.

## 2.2 Related Work

**Classical SSGs and attacker uncertainty.** The optimal commitment for a known attacker can be computed by an LP [5]. Uncertainty about the attacker has been handled by *robust optimization*, which plans against worst-case payoffs [18], and by *Bayesian SSGs*, which require a prior over attacker types [12, 15]. Both approaches rest on priors or constraints that are often hard to validate. Letchford et al. [13] give a query-efficient learning algorithm for a *single* fixed but unknown type, and Blum et al. [3] extend this to a setting that adapts round-by-round to one fixed type.

**Online learning against adversarial attacker sequences.** Balcan et al. [2] pivot from the single-type paradigm by allowing the sequence of attackers to be adversarially chosen from a known finite set  $K$ . For full information they give a polynomial-weights algorithm with regret  $O(\sqrt{Tn^2k\log(nk)})$  that partitions the coverage polytope into best-response regions and runs experts over the  $\epsilon$ -approximate extreme points; for bandit feedback they use a barycentric spanner construction to give an  $O(T^{2/3})$  bound. Crucially, they also prove a linear lower bound when  $|K|$  is unrestricted, which forces any non-trivial extension to restrict the attacker type set somehow. We take the weakest restriction: a known cardinality bound  $K_{\max}$ .

**Online learning with changing expert sets.** The sub-routine inside GROW-CATALOGUE is an anytime variant of Polynomial Weights / Hedge [4, 6, 14]. We restart this sub-routine each time the catalogue grows; simpler algorithms for tracking a growing set of experts exist [16], and for tracking a shrinking set [7], but both would only shave lower-order logarithmic factors.

**Other recent directions.** Haghtalab et al. [8] analyze no-regret learning against strategic (non-myopic) attackers in a single-type query model. Harris et al. [9] extend to a contextual Stackelberg setting in which both players observe side information. Neither addresses the *unknown  $K$  with bounded cardinality* question that we pose here.

## 3 Model

We adopt the repeated-SSG notation of Balcan et al. [2, Sec. 2]; our notation is compatible with theirs.

### 3.1 Game primitives

The game has a finite horizon  $T \in \mathbb{N}$  and  $n$  targets  $N = \{1, \dots, n\}$ . The defender has  $R$  resources and a collection  $\mathcal{D} \subseteq 2^N$  of *schedules*, where each schedule is a subset of targets that a single resource can defend simultaneously. An *assignment function*  $A : R \rightarrow 2^{\mathcal{D}}$  records the schedules that each resource may be assigned to. A *pure strategy* is any valid resource-to-schedule assignment; let  $m$  denote the number of pure strategies. A *mixed strategy* is a distribution  $\mathbf{s} \in \Delta_m$  over pure strategies, and its induced *coverage probability vector* is  $\mathbf{p} = M\mathbf{s} \in [0, 1]^n$ , where  $M \in \{0, 1\}^{n \times m}$  has  $M_{ij} = 1$  iff the  $j$ -th pure strategy covers target  $i$ . Write  $\mathcal{P}$  for the set of

valid coverage probability vectors; Every coverage vector  $\mathbf{p}$  is realizable by some mixed strategy, and conversely every mixed strategy induces a unique  $\mathbf{p} \in \mathcal{P}$ .

**Defender payoffs.** For each target  $i$ , the defender has a payoff  $u_d^c(i) \in [-1, 1]$  if  $i$  is attacked *and* covered, and a payoff  $u_d^u(i) \in [-1, 1]$  if  $i$  is attacked *and uncovered*. The defender's expected utility when target  $i$  is attacked under coverage  $\mathbf{p}$  is therefore

$$U_d(i, \mathbf{p}) = u_d^c(i)p_i + u_d^u(i)(1 - p_i), \quad (1)$$

which is *linear* in  $\mathbf{p}$  for each fixed target  $i$ .

### 3.2 Attacker types and best response

An attacker *type*  $\alpha$  is a pair  $(u_\alpha^c, u_\alpha^u) \in [-1, 1]^n \times [-1, 1]^n$  of per-target utilities following the same “covered/uncovered” convention as the defender. Its expected utility for attacking target  $i$  under coverage  $\mathbf{p}$  is  $U_\alpha(i, \mathbf{p}) = u_\alpha^c(i)p_i + u_\alpha^u(i)(1 - p_i)$ . Observing  $\mathbf{p}$ , type  $\alpha$  *best-responds*:

$$b_\alpha(\mathbf{p}) = \arg \max_{i \in N} U_\alpha(i, \mathbf{p}), \quad (2)$$

with ties broken arbitrarily but consistently. Note that  $b_\alpha$  is piecewise constant in  $\mathbf{p}$ , which makes the defender's realized payoff  $U_d(b_\alpha(\mathbf{p}), \mathbf{p})$  a discontinuous function of  $\mathbf{p}$ . This non-continuity is the central technical difficulty that distinguishes SSGs from vanilla online linear optimization, and it is what drives the polytope-based machinery of Balcan et al. [2].

### 3.3 Repeated game and regret

There is a (possibly unknown) set  $K$  of attacker types. An *oblivious* adversary chooses a sequence  $\mathbf{a} = a_1, \dots, a_T \in K^T$  before the game starts; the sequence is fixed but unknown to the defender. At each round  $t \in \{1, \dots, T\}$  the defender commits  $\mathbf{p}_t \in \mathcal{P}$ , attacker  $a_t$  plays  $b_{a_t}(\mathbf{p}_t)$ , and the defender receives the random payoff  $U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t)$ . We assume *full-information feedback*: at the end of round  $t$  the defender observes the identity of  $a_t$ , and the first time any type  $\alpha$  is observed, its utility functions are revealed as well.

The defender's benchmark is the best fixed coverage vector in hindsight:

$$\mathbf{p}^* = \arg \max_{\mathbf{p} \in \mathcal{P}} \sum_{t=1}^T U_d(b_{a_t}(\mathbf{p}), \mathbf{p}). \quad (3)$$

The expected regret of an online algorithm that plays  $\mathbf{p}_t$  at round  $t$  is

$$\text{Regret}(T) = \sum_{t=1}^T U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*) - \mathbb{E} \left[ \sum_{t=1}^T U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t) \right], \quad (4)$$

where the expectation is over the algorithm's internal randomization.

### 3.4 Preliminaries: polytope partition and $\mathcal{E}(C; \epsilon)$

The following definitions, taken from Balcan et al. [2, Sec. 4], are essential. For any catalogue  $C \subseteq K$ , a type  $\alpha \in C$ , and a target  $i \in N$ , the region in which  $\alpha$ 's best response is  $i$  is

$$\mathcal{P}_i^\alpha(C) = \{\mathbf{p} \in \mathcal{P} : b_\alpha(\mathbf{p}) = i\}, \quad (5)$$

which is the intersection of  $\mathcal{P}$  with the  $n - 1$  linear inequalities  $U_\alpha(i, \mathbf{p}) \geq U_\alpha(j, \mathbf{p})$  for  $j \neq i$ ; hence a convex polytope. For a *response profile*  $\sigma : C \rightarrow N$  we write

$$\mathcal{P}_\sigma(C) = \bigcap_{\alpha \in C} \mathcal{P}_{\sigma(\alpha)}^\alpha(C). \quad (6)$$

The non-empty sets  $\mathcal{P}_\sigma(C)$  form a partition of  $\mathcal{P}$  into convex regions in which the best response of every type in  $C$  is held fixed; within any such region, the defender's payoff under a fixed attacker type is a linear function of  $\mathbf{p}$ .

We will need three results from Balcan et al. [2], restated below in our notation and cited explicitly at every use. Lemmas 1 and 2 describe the approximate extreme-point set; Proposition 3 is the standard Polynomial-Weights online learning guarantee, adapted to the present loss range.

**Lemma 1** (2, Lemma 4.3). *For every  $\epsilon > 0$  and every catalogue  $C \subseteq K$ , there is a finite set  $\mathcal{E}(C; \epsilon) \subseteq \mathcal{P}$  of points, called the  $\epsilon$ -approximate extreme points of the partition, such that for any sequence  $\mathbf{a}' = a'_1, \dots, a'_T$ , with  $a'_t \in C$  for all  $t$ ,*

$$\max_{\mathbf{p} \in \mathcal{E}(C; \epsilon)} \sum_{t=1}^{T'} U_d(b_{a'_t}(\mathbf{p}), \mathbf{p}) \geq \max_{\mathbf{p} \in \mathcal{P}} \sum_{t=1}^{T'} U_d(b_{a'_t}(\mathbf{p}), \mathbf{p}) - 2\epsilon T'.$$

**Lemma 2** (2, Lemma 4.4). *The set  $\mathcal{E}(C; \epsilon)$  of Lemma 1 can be chosen so that  $|\mathcal{E}(C; \epsilon)| \in O((2^n + |C|n^2)^n \cdot n^{|C|})$ .*

**Proposition 3** (2, Proposition 1). *For any fixed expert set  $\mathcal{M}$  and any sequence of losses in  $[-\kappa, \kappa]$ , there is an online learning algorithm (an anytime variant of Polynomial Weights / Hedge; see 4, Sec. 2) whose expected regret after  $T'$  rounds is at most  $O(\sqrt{T'} \kappa \log |\mathcal{M}|)$ , without requiring  $T'$  to be known in advance.*

## 4 The Unknown- $K$ Setting

We now state the problem that our algorithm and theorem address.

**Problem 4** (Unknown- $K$  repeated SSG). A repeated SSG as in Section 3 is played, with two modifications.

1. The set  $K$  of attacker types is not revealed to the defender at the start; however, the defender is given a constant  $K_{\max} \geq 1$  with the promise  $|K| \leq K_{\max}$ .
2. Feedback is full information: after round  $t$  the defender observes  $a_t$ , and the first time any type  $\alpha$  appears, its utility vectors  $(u_\alpha^c, u_\alpha^u)$  are revealed.

The benchmark  $\mathbf{p}^*$  and regret  $\text{Regret}(T)$  are as in Equations (3) and (4).

**Remark 5** (Necessity of the bound  $K_{\max}$ ). Balcan et al. [2, Theorem 7.1] exhibit, for every  $T$ , a game with  $|K| < 2^{T+1}$  on which every online algorithm has expected regret at least  $T/2$ . Hence some finite bound on  $|K|$  is *necessary* for sublinear regret; Problem 4 takes the weakest such bound, namely a known constant  $K_{\max}$ .

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**Algorithm 1** GROW-CATALOGUE (full information).

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**Require:** horizon  $T$ , bound  $K_{\max}$ , game primitives, default  $\mathbf{p}_{\text{def}} \in \mathcal{P}$

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1:  $C \leftarrow \emptyset$ ;  $\mathcal{E} \leftarrow \emptyset$ ;  $\epsilon \leftarrow 1/\sqrt{T}$ ;  $t_{\text{loc}} \leftarrow 0$ 
2: for  $t = 1, \dots, T$  do
3:   if  $\mathcal{E} = \emptyset$  then
4:     play  $\mathbf{p}_t \leftarrow \mathbf{p}_{\text{def}}$ 
5:   else
6:     sample  $\mathbf{p}_t \sim q_t$ , where  $q_t$  is the current PW distribution
   over  $\mathcal{E}$ 
7:   end if
8:   observe  $a_t$ 
9:   if  $a_t \notin C$  then  $\triangleright$  discovery round: new attacker type
10:     $C \leftarrow C \cup \{a_t\}$ ; record  $(u_{a_t}^c, u_{a_t}^u)$ 
11:     $\mathcal{E} \leftarrow \mathcal{E}(C; \epsilon)$   $\triangleright$  recompute extreme points
12:    reset PW on  $\mathcal{E}$ ;  $t_{\text{loc}} \leftarrow 0$ 
13:   else  $\triangleright$  routine round
14:      $t_{\text{loc}} \leftarrow t_{\text{loc}} + 1$ 
15:     compute  $\ell_t(\mathbf{p}) = -U_d(b_{a_t}(\mathbf{p}), \mathbf{p})$  for  $\mathbf{p} \in \mathcal{E}$ 
16:      $\eta \leftarrow \sqrt{\log |\mathcal{E}| / t_{\text{loc}}}$ 
17:      $w_{\mathbf{p}} \leftarrow w_{\mathbf{p}} \cdot e^{-\eta \ell_t(\mathbf{p})}$  for all  $\mathbf{p} \in \mathcal{E}$ 
18:   end if
19: end for
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## 5 Algorithm and Main Result

### 5.1 The Grow-Catalogue algorithm

We describe GROW-CATALOGUE at a high level and then give the pseudocode in Algorithm 1. At any round  $t$  the defender holds a catalogue  $C_t \subseteq K$  of all types seen so far. Conceptually, the defender runs a small instance of the *known- $K$*  algorithm of Balcan et al. [2, Sec. 5] on the catalogue  $C_t$ , pretending that  $C_t$  is the full attacker set. Concretely:

- The algorithm maintains the  $\epsilon$ -approximate extreme-point set  $\mathcal{E}_t = \mathcal{E}(C_t; \epsilon)$  of Lemma 1. We set  $\epsilon = 1/\sqrt{T}$ ; this balances the  $2\epsilon T$  approximation term in Lemma 1 against the  $\sqrt{T \log N_{\max}}$  regret term of Proposition 3.
- An anytime Polynomial-Weights sub-routine of Proposition 3 runs over  $\mathcal{E}_t$ , updating via a time-varying learning rate  $\eta_{t_{\text{loc}}} = \sqrt{\log |\mathcal{E}_t| / t_{\text{loc}}}$  where  $t_{\text{loc}}$  is a local counter reset at every catalogue change. This is the standard anytime schedule; see Cesa-Bianchi and Lugosi [4, Sec. 2].
- When a new attacker type  $\alpha$  appears, the catalogue grows to  $C_t \cup \{\alpha\}$ , the approximate extreme-point set is recomputed, and the sub-routine is restarted on the enlarged expert set. The single round of first contact with  $\alpha$  contributes a bounded additive amount to the regret that we will absorb.

### 5.2 Main Result

**Theorem 6 (Main).** *Let the defender play Algorithm 1 in the unknown- $K$  full-information setting of Problem 4. For any oblivious adversarial sequence  $\mathbf{a} \in K^T$  with  $|K| \leq K_{\max}$ , the expected regret satisfies*

$$\mathbb{E}[\text{Regret}(T)] \leq O\left(nK_{\max}\sqrt{T \log(nK_{\max})}\right). \quad (7)$$

*In particular, if  $K_{\max} = O(1)$  then  $\mathbb{E}[\text{Regret}(T)] = \tilde{O}(n\sqrt{T})$ , matching the known- $K$  bound of Balcan et al. [2, Theorem 5.1] up to logarithmic factors.*

We prove Theorem 6 in detail in Section 6. Two useful corollaries of the proof (the regime of applicability and an instance-adaptive refinement) are stated after the proof in Section 6.1.

## 6 Proof of Theorem 6

The proof closely tracks the structure of the known- $K$  proof of Balcan et al. [2, Sec. 5] but has two non-trivial deviations: (i) the expert set is no longer static but grows over time, and (ii) the discovery rounds introduce an extra bounded additive term. We account for both the deviations in our proof.

*Proof:-* Fix an oblivious adversarial sequence  $\mathbf{a} = a_1, \dots, a_T \in K^T$  with  $|K| \leq K_{\max}$ . Let  $K' = |\{a_1, \dots, a_T\}| \leq K_{\max}$  denote the number of distinct types that actually appear in  $\mathbf{a}$ . The proof proceeds in ten steps.

**Step 1 (Epoch decomposition).** Let  $\tau_1 < \tau_2 < \dots < \tau_{K'}$  be the round indices at which each new attacker type is first observed, i.e., for each  $i \in \{1, \dots, K'\}$ , round  $\tau_i$  is the smallest  $t$  such that  $a_t \notin \{a_1, \dots, a_{t-1}\}$ . We define the *epochs*

$$\text{Ep}_i = \{\tau_i + 1, \tau_i + 2, \dots, \tau_{i+1} - 1\}, \quad i = 1, \dots, K', \quad (8)$$

with the convention  $\tau_{K'+1} := T + 1$ , and write  $L_i = |\text{Ep}_i|$ . Let  $\mathcal{D} = \{\tau_1, \dots, \tau_{K'}\}$  be the set of *discovery rounds*. By construction,

$$[1, T] = \mathcal{D} \sqcup \bigcup_{i=1}^{K'} \text{Ep}_i,$$

and  $|\mathcal{D}| + \sum_{i=1}^{K'} L_i = T$  with  $|\mathcal{D}| = K' \leq K_{\max}$ . The epoch decomposition is the critical structural property of the proof: it re-interprets the online problem with a growing expert set as a finite sequence of online problems with *fixed* expert sets, so that the known- $K$  analysis of Balcan et al. [2] can be invoked inside each epoch.

**Step 2 (Catalogue is constant inside an epoch).** Immediately after round  $\tau_i$ , the catalogue maintained by Algorithm 1 is  $C_i = \{a_{\tau_1}, \dots, a_{\tau_i}\}$ , and it is not updated until the next discovery round  $\tau_{i+1}$ . Hence  $C_t = C_i$  for every  $t \in \text{Ep}_i$ . Moreover, by the defining property of  $\tau_{i+1}$ , no new type appears in  $\text{Ep}_i$ , so

$$a_t \in C_i \quad \text{for every } t \in \text{Ep}_i. \quad (9)$$

Denote by  $\mathcal{E}_i := \mathcal{E}(C_i; \epsilon)$  the extreme-point set used by the PW sub-routine during  $\text{Ep}_i$ ; Equation (9) is exactly the hypothesis needed to apply Lemma 1 to  $C_i$  and the sub-sequence inside  $\text{Ep}_i$ .

**Step 3 (Within-epoch regret).** During  $\text{Ep}_i$  the algorithm runs a fresh instance of the anytime Polynomial-Weights sub-routine on the fixed expert set  $\mathcal{E}_i$  for  $L_i$  rounds. The losses are  $\ell_t(\mathbf{p}) = -U_d(b_{a_t}(\mathbf{p}), \mathbf{p}) \in [-1, 1]$ , so the range parameter is  $\kappa = 1$ . Proposition 3 therefore yields an absolute constant

$c_1 > 0$  such that

$$\max_{\mathbf{p} \in \mathcal{E}_i} \sum_{t \in \text{Ep}_i} U_d(b_{a_t}(\mathbf{p}), \mathbf{p}) - \mathbb{E} \left[ \sum_{t \in \text{Ep}_i} U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t) \right] \leq c_1 \sqrt{L_i \log |\mathcal{E}_i|}. \quad (10)$$

**Step 4 (Within-epoch approximation via Lemma 1).** We now wish to compare the max over  $\mathcal{E}_i$  during  $\text{Ep}_i$  with the *global* benchmark  $\mathbf{p}^*$  (the best fixed strategy over the full horizon  $[1, T]$ ). Lemma 1, however, is a statement of the form “ $\max_{\mathbf{p} \in \mathcal{E}(C; \epsilon)}(\cdot) \geq \max_{\mathbf{p} \in \mathcal{P}}(\cdot) - 2\epsilon T'$ ” and, strictly speaking, does *not* directly involve  $\mathbf{p}^*$ : it compares the max over the finite set  $\mathcal{E}(C; \epsilon)$  against the unrestricted max over  $\mathcal{P}$ , both on the *same* attacker sequence. We therefore proceed in two sub-steps and make the passage to  $\mathbf{p}^*$  explicit.

*Sub-step 4a (apply Lemma 1 to the epoch sub-sequence).* By Equation (9), the sub-sequence  $(a_t)_{t \in \text{Ep}_i}$  has length  $L_i$  and uses only types in  $C_i$ , which is exactly the hypothesis of Lemma 1. Applying the lemma with catalogue  $C_i$  and sequence length  $T' = L_i$  gives

$$\begin{aligned} \max_{\mathbf{p} \in \mathcal{E}_i} \sum_{t \in \text{Ep}_i} U_d(b_{a_t}(\mathbf{p}), \mathbf{p}) \\ \geq \max_{\mathbf{p} \in \mathcal{P}} \underbrace{\sum_{t \in \text{Ep}_i} U_d(b_{a_t}(\mathbf{p}), \mathbf{p})}_{=: \text{OPT}_i} - 2\epsilon L_i. \end{aligned} \quad (11)$$

Here  $\text{OPT}_i$  is the best cumulative utility achievable on  $\text{Ep}_i$  by *any* coverage vector in  $\mathcal{P}$ ; crucially, this is the epoch- $i$  optimum, which in general *differs* from the value of  $\mathbf{p}^*$ . This is the subtle point that distinguishes our setting from the known- $K$  proof of Balcan et al. [2, Sec. 5]: in their analysis, Lemma 1 is applied once on the full horizon, so  $\text{OPT} = \sum_{t=1}^T U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*)$  by definition of  $\mathbf{p}^*$  and no separate passage is needed. In our per-epoch application,  $\text{OPT}_i$  is an epoch-local quantity, potentially larger than  $\mathbf{p}^*$ ’s value on the epoch.

*Sub-step 4b (pass from  $\text{OPT}_i$  to  $\mathbf{p}^*$ ).* Since  $\mathbf{p}^* \in \mathcal{P}$ , the coverage vector  $\mathbf{p}^*$  is a *feasible* competitor in the maximization defining  $\text{OPT}_i$ . Therefore

$$\text{OPT}_i \geq \sum_{t \in \text{Ep}_i} U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*). \quad (12)$$

Inequality Equation (12) is *not* a consequence of Lemma 1 but of the trivial fact that the global benchmark  $\mathbf{p}^*$  is an element of the set over which the epoch-local optimum is taken. Chaining Equation (11) with Equation (12),

$$\begin{aligned} \max_{\mathbf{p} \in \mathcal{E}_i} \sum_{t \in \text{Ep}_i} U_d(b_{a_t}(\mathbf{p}), \mathbf{p}) \\ \geq \sum_{t \in \text{Ep}_i} U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*) - 2\epsilon L_i. \end{aligned} \quad (13)$$

**Step 5 (Per-epoch combined bound).** Chaining Equations (10) and (13), for every epoch  $i \in \{1, \dots, K'\}$  we obtain

$$\begin{aligned} \mathbb{E} \left[ \sum_{t \in \text{Ep}_i} U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t) \right] \\ \geq \sum_{t \in \text{Ep}_i} U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*) - 2\epsilon L_i - c_1 \sqrt{L_i \log |\mathcal{E}_i|}. \end{aligned} \quad (14)$$

Equation (14) is the per-epoch regret bound *against the global benchmark*  $\mathbf{p}^*$ , and it is the building block for the rest of the proof. Note that despite the fact that  $\mathbf{p}^*$  is the benchmark over all  $T$  rounds (not just epoch  $i$ ), we are able to bound our per-epoch regret against  $\mathbf{p}^*$  because  $\mathbf{p}^* \in \mathcal{P}$  and we applied Lemma 1 on the sub-sequence.

**Step 6 (Discovery rounds).** For each discovery round  $t \in \mathcal{D}$ , the algorithm either plays  $\mathbf{p}_{\text{def}}$  (if  $t = \tau_1$ ) or samples  $\mathbf{p}_t$  from the PW distribution inherited from  $\text{Ep}_{i-1}$  (if  $t = \tau_i$  for  $i \geq 2$ ). In both cases the sampled  $\mathbf{p}_t$  is not tuned to the newly-appearing type  $a_t = a_{\tau_i}$ . However, because the payoff  $U_d$  is bounded in  $[-1, 1]$ , for every  $\mathbf{p} \in \mathcal{P}$  and every type  $\alpha$ ,  $U_d(b_\alpha(\mathbf{p}), \mathbf{p}) \in [-1, 1]$ , so

$$|U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*) - U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t)| \leq 2. \quad (15)$$

Summing Equation (15) over the at most  $K' \leq K_{\max}$  discovery rounds yields

$$\sum_{t \in \mathcal{D}} U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*) - \mathbb{E} \left[ \sum_{t \in \mathcal{D}} U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t) \right] \leq 2K_{\max}. \quad (16)$$

This is the only source of additive dependence on  $K_{\max}$  that does not come from the main  $\sqrt{TK_{\max}}$  term. It is the “price of not knowing  $K$ ” at a discovery round.

**Step 7 (Uniform bound on  $|\mathcal{E}_i|$ ).** By Lemma 2 applied with  $|C_i| \leq K_{\max}$ ,

$$|\mathcal{E}_i| \leq c_2 (2^n + K_{\max} n^2)^n \cdot n^{K_{\max}} =: N_{\max},$$

for an absolute constant  $c_2$ . Taking logarithms,

$$\begin{aligned} \log N_{\max} &= n \log(2^n + K_{\max} n^2) + K_{\max} \log n + O(1) \\ &\leq n(n + 1 + \log(K_{\max} n^2)) + K_{\max} \log n + O(1) \\ &= O(n^2 + n \log(n K_{\max}) + K_{\max} \log n) \\ &= O(n^2 K_{\max} \log(n K_{\max})). \end{aligned} \quad (17)$$

Both summands of  $2^n + K_{\max} n^2$  are at least 1, so their sum is at most  $2 \cdot 2^n \cdot K_{\max} n^2$ , which gives the second line; the last step absorbs each of the three terms into the product, every factor of which is at least 1 once  $n \geq 2$ . The third line is sharper and could be carried through; we quote the fourth because it is the form Balcan et al. [2, Lemma 4.4] take into their Theorem 5.1, which lets Theorem 6 be read off against that theorem exactly. Crucially, Equation (17) is uniform in  $i$ :  $\log |\mathcal{E}_i| \leq \log N_{\max}$  for every epoch. This lets us pull the  $\log |\mathcal{E}_i|$  factor in Step 5 outside the sum over epochs.

**Step 8 (Summing across epochs).** Sum Equation (14) over  $i = 1, \dots, K'$  and add Equation (16). The algorithm's total expected utility is then at least the benchmark's total utility, minus the discovery term  $2K_{\max}$ , minus  $2\epsilon T$  from the telescoping Lemma-4.3 approximation errors (using  $\sum_i L_i \leq T$ ), minus the sum of the per-epoch PW regret terms. Rearranging,

$$\mathbb{E}[\text{Regret}(T)] \leq 2K_{\max} + 2\epsilon T + c_1 \sqrt{\log N_{\max}} \sum_{i=1}^{K'} \sqrt{L_i}. \quad (18)$$

**Step 9 (Cauchy–Schwarz).** The last term involves the sum of  $K'$  square roots. By the Cauchy–Schwarz inequality applied to  $(1, \dots, 1) \in \mathbb{R}^{K'}$  and  $(\sqrt{L_1}, \dots, \sqrt{L_{K'}})$ ,

$$\sum_{i=1}^{K'} \sqrt{L_i} \leq \sqrt{K' \cdot \sum_{i=1}^{K'} L_i} \leq \sqrt{K_{\max} T}. \quad (19)$$

The first inequality is tight when all epochs have equal length, a scenario that corresponds to an adversary who reveals types evenly across the horizon. This is where the leading multiplicative  $\sqrt{K_{\max}}$  factor in the bound is generated. An instance-adaptive version uses  $K'$  in place of  $K_{\max}$ , which is what Corollary 8 records.

**Step 10 (Final substitution).** Plugging Equations (17) and (19) into Equation (18) and choosing  $\epsilon = 1/\sqrt{T}$  gives

$$\begin{aligned} \mathbb{E}[\text{Regret}(T)] &\leq 2K_{\max} + 2\sqrt{T} + c_1 \sqrt{K_{\max} T \log N_{\max}} \\ &\leq 2K_{\max} + 2\sqrt{T} \\ &\quad + c_3 n K_{\max} \sqrt{T \log(nK_{\max})} \end{aligned}$$

for an absolute constant  $c_3$ , the last line by substituting Equation (17) for  $\log N_{\max}$  and taking the square root of the product, which collects the two factors of  $K_{\max}$  into one. Whenever  $T \geq K_{\max}$  (the only regime in which Theorem 6 is non-trivial) the leading term dominates the additive  $2K_{\max} + 2\sqrt{T}$ , so

$$\mathbb{E}[\text{Regret}(T)] = O\left(nK_{\max} \sqrt{T \log(nK_{\max})}\right),$$

which is the bound of Theorem 6. Written in the equivalent form  $O(\sqrt{T n^2 K_{\max}^2 \log(nK_{\max})})$ , it is the bound of Balcan et al. [2, Theorem 5.1] with  $k$  replaced by  $K_{\max}$  and inflated by the  $\sqrt{K_{\max}}$  of the Cauchy–Schwarz step, so Step 9 is where the price of the unknown catalogue is paid. This completes the proof.  $\square$

## 6.1 Consequences of the proof

Two corollaries follow directly from the proof of Theorem 6 above. The first characterizes the regime in which the bound is non-trivial, the second gives an instance-adaptive refinement.

**Corollary 7** (Regime of applicability). *The bound of Theorem 6 is sublinear in  $T$  if and only if  $n^2 K_{\max}^2 \log(nK_{\max}) = o(T)$ ; equivalently,  $K_{\max} = o(\sqrt{T}/(n\sqrt{\log(nK_{\max})}))$ , which up to the logarithmic factor is  $K_{\max} = \tilde{o}(\sqrt{T}/n)$ . In particular, for  $K_{\max} = O(1)$  the rate is  $\tilde{O}(n\sqrt{T})$ .*

*Proof.* Set the right-hand side of Equation (7) equal to  $T$  and solve for  $K_{\max}$ . Squaring,  $nK_{\max} \sqrt{T \log(nK_{\max})} = o(T)$  holds exactly when  $n^2 K_{\max}^2 \log(nK_{\max}) = o(T)$ , which is the stated condition; dividing by  $n^2 \log(nK_{\max})$  and taking square roots gives the equivalent form.  $\square$

**Corollary 8** (Instance-adaptive bound). *Let  $K' = |\{a_1, \dots, a_T\}| \leq K_{\max}$  be the number of distinct attacker types actually observed in the realized sequence. Then Algorithm 1 achieves*

$$\mathbb{E}[\text{Regret}(T)] \leq O\left(nK' \sqrt{T \log(nK')}\right).$$

*Proof.*  $K'$  replaces  $K_{\max}$  at both places the proof uses it: the Cauchy–Schwarz step Equation (19) gives  $\sum_{i=1}^{K'} \sqrt{L_i} \leq \sqrt{K' T}$ , and the catalogue never holds more than the types observed, so  $|C_i| \leq K'$  and Step 7 gives  $\log N_{\max} = O(n^2 K' \log(nK'))$ . Redoing Step 10 with both gives the bound.  $\square$

## 7 Partial-Information Feedback with an Unknown Type Set

Theorem 6 shows that not knowing  $K$  is affordable: under full-information feedback it costs exactly a factor  $\sqrt{K_{\max}}$  over the known- $K$  rate of Balcan et al. [2, Theorem 5.1], as Step 10 of Section 6 records. Not observing  $a_t$  is affordable too: with  $K$  known and only the attacked target observed, Balcan et al. [2, Theorem 6.1] achieve  $O(T^{2/3} n k \log^{1/3}(nk))$ . One expects the two relaxations to compose. They do not.

We show that under partial information with an unknown type set no algorithm achieves regret  $o(T)$ , even when  $K_{\max} = 1$ , so that one fixed type attacks in every round. The obstruction is not how many types the adversary has, but that the defender can never learn *which* type it faces: it sees only a target index, and infinitely many types produce that same index. The diagnosis is that the defender is missing a *name*, and a name is a bounded object, which is what Section 8 then supplies. Proposition 14 closes the section by showing that this naming account is special to  $K_{\max} = 1$ : two types already force  $\Omega(\sqrt{T})$  on a two-element universe, under full information and with  $K$  known.

### 7.1 The problem

**Problem 9** (Unknown- $K$  repeated SSG, partial information). A repeated SSG as in Section 3 is played, with three modifications.

1. There is a set  $\mathcal{U}$  of attacker types, the *universe*, known to the defender and not assumed finite.
2. The set  $K \subseteq \mathcal{U}$  is not revealed to the defender; it is given only a constant  $K_{\max} \geq 1$  with the promise  $|K| \leq K_{\max}$ .
3. Feedback is *partial information* [2, Section 3]: at the end of round  $t$  the defender observes only the attacked target  $b_{a_t}(\mathbf{p}_t)$ . It never observes  $a_t$ , and utility vectors are never revealed.

The adversary is oblivious, and  $\mathbf{p}^*$  and  $\text{Regret}(T)$  are as in Equations (3) and (4).

The defender's own payoff is not extra feedback: it knows  $u_d^c$  and  $u_d^u$ , so  $U_d(i_t, \mathbf{p}_t)$  follows from the observed target and its own play. And Problem 9 is at least as hard as each setting it weakens: revealing  $a_t$  recovers Problem 4, and revealing  $K$  at  $t = 0$  recovers Balcan et al. [2, Section 6].

## 7.2 The gadget $G_3$

Every lower bound below is proved in one fixed game  $G_3$ : the game of Balcan et al. [2, Theorem 7.1], with  $n = 3$  targets, one resource, and schedules  $\mathcal{D} = \{\emptyset, \{1\}, \{2\}, \{3\}\}$ , so that  $\mathcal{P} = \{\mathbf{p} \in [0, 1]^3 : p_1 + p_2 + p_3 \leq 1\}$ . Every coverage vector we exhibit has  $\sum_i p_i = 1$ , so the arguments run verbatim on the simplex as well. The defender's payoffs are

$$u_d^c(i) = u_d^u(i) = -1 \quad (i \in \{1, 3\}), \quad u_d^c(2) = u_d^u(2) = 0, \quad (20)$$

so that by Equation (1)

$$U_d(1, \mathbf{p}) = U_d(3, \mathbf{p}) = -1, \quad U_d(2, \mathbf{p}) = 0 \quad (21)$$

for every  $\mathbf{p} \in \mathcal{P}$ . The defender's payoff therefore depends on  $\mathbf{p}_t$  only through which target the attacker picks: a round is worth 0 if target 2 is attacked and  $-1$  otherwise. We say the defender *hits* at round  $t$  if  $b_{a_t}(\mathbf{p}_t) = 2$  and *misses* otherwise. Ties go to the lowest-indexed target, as in Balcan et al. [2, Section 7], though target 2 is the *strict* maximizer at every benchmark we exhibit, so no bound below depends on the rule.

**Threshold types.** For  $0 \leq r_1 \leq r_2 \leq 1$  let  $\alpha_{r_1, r_2}$  be the attacker with

$$u_\alpha^c = (r_1 - 1, 0, -r_2), \quad u_\alpha^u = (r_1, 0, 1 - r_2), \quad (22)$$

all six entries of which lie in  $[-1, 1]$ . Substituting into  $U_\alpha(i, \mathbf{p}) = u_\alpha^c(i)p_i + u_\alpha^u(i)(1 - p_i)$  gives, for every  $\mathbf{p} \in \mathcal{P}$ ,

$$U_\alpha(1, \mathbf{p}) = r_1 - p_1, \quad U_\alpha(2, \mathbf{p}) = 0, \quad U_\alpha(3, \mathbf{p}) = (1 - r_2) - p_3. \quad (23)$$

So on the segment  $\mathbf{p} = (p, 0, 1 - p)$  the three utilities are  $r_1 - p$ , 0 and  $p - r_2$ , and comparing them gives

$$b_{\alpha_{r_1, r_2}}((p, 0, 1 - p)) = \begin{cases} 1 & p \in [0, r_1], \\ 2 & p \in (r_1, r_2], \\ 3 & p \in (r_2, 1]. \end{cases} \quad (24)$$

This is Balcan et al. [2, Lemma 7.2], with the utility vectors recorded explicitly because later arguments need the best response on all of  $\mathcal{P}$ , not only on this segment. A type is thus named by an interval  $(r_1, r_2]$ , and the defender's task is to locate it.

The next lemma is the structural fact driving every lower bound: a hit is very informative but very rare, because one coverage vector can hit at most one type of any disjoint family.

**Lemma 10** (the hit condition). *Let  $0 \leq r_1 < r_2 \leq 1$  and  $\alpha = \alpha_{r_1, r_2}$ . For every  $\mathbf{p} \in \mathcal{P}$ ,*

$$b_\alpha(\mathbf{p}) = 2 \iff p_1 > r_1 \text{ and } p_3 \geq 1 - r_2, \quad (25)$$

*and then  $p_1 \in (r_1, r_2]$ . Hence if the intervals of a family of types are pairwise disjoint, every  $\mathbf{p} \in \mathcal{P}$  hits at most one of them.*

*Proof.* By Equation (23) and lowest-index tie-breaking, target 2 is chosen exactly when  $U_\alpha(2, \mathbf{p}) = 0$  strictly beats  $U_\alpha(1, \mathbf{p}) = r_1 - p_1$  and weakly beats  $U_\alpha(3, \mathbf{p}) = (1 - r_2) - p_3$ , which is Equation (25). Then  $p_1 \leq 1 - p_2 - p_3 \leq 1 - p_3 \leq r_2$ , so  $p_1 \in (r_1, r_2]$ ; and a single  $p_1$  cannot lie in two disjoint intervals.  $\square$

## 7.3 Identification is a comparison search

Fix  $M \geq 2$  and let

$$I_j = \left( \frac{j-1}{M}, \frac{j}{M} \right], \quad \alpha^{(j)} = \alpha_{\frac{j-1}{M}, \frac{j}{M}}, \quad j \in [M], \quad (26)$$

so  $I_1, \dots, I_M$  partition  $(0, 1]$  and  $\mathcal{U}_M := \{\alpha^{(1)}, \dots, \alpha^{(M)}\}$  consists of  $M$  distinct types, to which Lemma 10 applies.

Facing an instance  $K = \{\alpha^{(j)}\}$ , the defender is searching for the index  $j$ : each play is a query, each observation an answer in  $\{1, 2, 3\}$ , and by Lemma 10 the answer “2” is consistent with at most one candidate. The following lemma bounds how fast *any* such search can succeed. It sharpens the classical  $\Omega(\log_2 M)$  comparison-search bound in the two ways we need: it charges every failed query rather than counting queries to termination, and it bounds the whole tail of the success time rather than its worst case alone.

**Lemma 11** (comparison trees identify slowly). *Let  $M \geq 1$  and let  $j$  be uniform on  $[M]$ . Consider any interactive protocol in which, for  $t = 1, 2, \dots$ , a deterministic strategy picks a query  $q_t$  as a function of everything seen so far and receives an answer  $o_t = \mathcal{O}(q_t, j) \in \{1, 2, 3\}$ . Assume only that*

$$|\{j' \in [M] : \mathcal{O}(q, j') = 2\}| \leq 1 \quad \text{for every query } q. \quad (27)$$

*Let  $\tau = \min\{t : o_t = 2\}$ , with  $\tau = \infty$  if there is no such  $t$ . Then for every  $u \geq 1$  and every  $T \geq 1$ ,*

$$\Pr[\tau \leq u] \leq \frac{2^u - 1}{M}, \quad (28)$$

$$\mathbb{E}[\min\{\tau - 1, T\}] \geq T - \frac{2^{T+1} - T - 2}{M}. \quad (29)$$

It is worth saying Equation (27) in words, since the whole bound rests on it: whatever the searcher asks, at most one of the  $M$  candidates would answer 2. So the answer 2 names the hidden index outright, and it is the only answer that does; 1 and 3 merely say “too low” or “too high” and narrow the field. This is the only property of the game the lemma uses, and Lemma 10 is what supplies it, which is why the same lemma serves again in Theorem 22.

*Proof.* Everything follows from counting, for each round  $t$ , how many of the  $M$  indices can be found for the first time at that round. Fix  $t$  and consider the indices with  $\tau(j) = t$ . Three facts combine.

(i) *There are at most  $2^{t-1}$  histories the searcher can be in.* Its only record of rounds  $1, \dots, t-1$  is the answers  $o_1, \dots, o_{t-1}$ , and none of them is 2, since  $\tau(j) = t$  means it has not hit yet. So each is 1 or 3, and the set  $H_t$  of such histories has  $|H_t| \leq 2^{t-1}$ .

(ii) *The history determines the round- $t$  query.* The strategy is deterministic, so that query is a fixed function  $q_t(h)$  of the history  $h$  alone, with no other input.

(iii) A query is answered 2 by at most one index. This is Equation (27).

Now take any  $j$  with  $\tau(j) = t$ . It produces some history  $h \in H_t$ , by (ii) the searcher then plays  $q_t(h)$ , and by definition of  $\tau$  the answer is 2; by (iii) at most one index does that. So each history accounts for at most one such  $j$ , and

$$|\{j : \tau(j) = t\}| \leq |H_t| \leq 2^{t-1}.$$

Summing over the first  $u$  rounds,

$$|\{j : \tau(j) \leq u\}| \leq \sum_{t=1}^u 2^{t-1} = 2^u - 1,$$

and dividing by  $M$ , since  $j$  is uniform on  $[M]$ , gives Equation (28).

For Equation (29), put  $X = \min\{\tau - 1, T\}$ . Since  $X > s$  iff  $\tau > s + 1$ ,

$$\begin{aligned} \mathbb{E}[X] &= \sum_{s=0}^{T-1} \Pr[X > s] = \sum_{u=1}^T (1 - \Pr[\tau \leq u]) \\ &\geq T - \sum_{u=1}^T \frac{2^u - 1}{M}, \end{aligned}$$

which is Equation (29) because  $\sum_{u \leq T} 2^u = 2^{T+1} - 2$ .  $\square$

It is worth seeing why Equation (29) is so close to  $T$ , since that is what makes Theorem 12 below cost all but a constant number of rounds. The census  $|\{j : \tau(j) = t\}| \leq 2^{t-1}$  leaves no slack: summed to  $t = T$  it is  $2^T - 1$ , so at  $M = 2^T$  the budget is exactly spent, and the best a search can do is to find  $2^{T-1}$  of the indices at round  $T$ ,  $2^{T-2}$  at round  $T - 1$ , and so on. Half the candidates are therefore pinned down only by the very last comparison, a quarter by the one before it. Binary search has this shape in general: its average depth is its worst-case depth minus a constant, not a fraction of it. That constant is the 2 in Equation (30).

## 7.4 The impossibility theorem

**Theorem 12** (unknown  $K$  with partial information is not learnable). *Let  $G_3$  be as in Equation (20) and let  $\mathfrak{U} = \{\alpha_{r_1, r_2} : 0 \leq r_1 \leq r_2 \leq 1\}$  be the infinite universe of threshold types Equation (22). For every  $T \geq 1$  and every  $K_{\max} \geq 1$ , every online algorithm for Problem 9, randomized or deterministic, even one knowing  $T$ ,  $\mathfrak{U}$ ,  $K_{\max}$  and the construction, suffers, on some instance with  $|K| = 1$ ,*

$$\mathbb{E}[\text{Regret}(T)] \geq T - 2 + \frac{T + 2}{2^T} > T - 2. \quad (30)$$

*The game is fixed; only the adversary's choice of  $K$  within it depends on  $T$ .*

*Proof.* Set  $M = 2^T$  and take  $\mathfrak{U}_M \subset \mathfrak{U}$  as in Equation (26). By Yao's minimax principle it suffices to exhibit a distribution over instances against which every deterministic algorithm suffers this much: draw  $j$  uniformly from  $[M]$  and set  $K = \{\alpha^{(j)}\}$ , with  $a_t = \alpha^{(j)}$  for all  $t$ . This is legal for every  $K_{\max} \geq 1$  and oblivious, since  $\mathbf{a}$  is fixed before play.

The benchmark is 0. Let  $\mathbf{p}^* = \left(\frac{2j-1}{2M}, 0, 1 - \frac{2j-1}{2M}\right)$ , whose first coordinate is the midpoint of  $I_j$ . Then  $p_1^* > r_1$  and

$p_3^* > 1 - r_2$  strictly, so  $b_{\alpha^{(j)}}(\mathbf{p}^*) = 2$  by Equation (25) under any tie-breaking rule, and  $\mathbf{p}^*$  earns 0 every round. As  $U_d \leq 0$  everywhere by Equation (21), no coverage vector does better, so  $\mathbf{p}^*$  is a maximizer in Equation (3) and

$$\text{Regret}(T) = \mathbb{E}[\#\{t \leq T : \text{miss at } t\}]. \quad (31)$$

The interaction is a comparison search. Observations are the algorithm's only feedback, so a deterministic one is a family of maps  $\mathbf{p}_t = \mathbf{p}_t(o_1, \dots, o_{t-1})$ . Take  $q_t = \mathbf{p}_t$  and  $\mathcal{O}(\mathbf{p}, j') = b_{\alpha^{(j')}}(\mathbf{p})$ ; the  $I_j$  are disjoint, so Lemma 10 gives exactly Equation (27).

*Counting.* Every round before the first hit  $\tau$  is a miss, so by Equation (31) and Equation (29) with  $M = 2^T$ ,

$$\mathbb{E}_j[\text{Regret}(T)] \geq \mathbb{E}_j[\min\{\tau - 1, T\}] \geq T - \frac{2^{T+1} - T - 2}{2^T},$$

which is Equation (30). Averaging over a randomized algorithm's internal randomness preserves the bound, since it holds for every deterministic realization.  $\square$

*Remark 13* (the bound is exactly tight). The constant cannot be improved: the defender that keeps the set of indices consistent with its observations, an integer interval, by Equation (24), and plays  $\mathbf{p}_t = (k/M, 0, 1 - k/M)$  for its midpoint  $k$  attains expected regret exactly  $T - 2 + (T + 2)2^{-T}$ , as Section 9.4 verifies. Since  $\text{Regret}(T) \leq T$  always in  $G_3$ , the defender loses all but two of the  $T$  rounds.

Letchford et al. [13] show that a defender facing a *single* unknown attacker type can learn a near-optimal commitment from a small number of best-response queries, which at first sight is in tension with Theorem 12. There is none: any such bound must depend on the geometry of the best-response regions, and the family  $\mathfrak{U}_M$  of Equation (26) drives the region that has to be located down to width  $1/M = 2^{-T}$ .

## 7.5 A $\sqrt{T}$ floor as soon as $K_{\max} \geq 2$

Is a name all the defender ever lacks? For  $K_{\max} = 1$ , yes: on a fixed finite universe  $\mathfrak{U}_M$  the version-space search of Remark 13 stops missing after  $\lceil \log_2 M \rceil$  rounds, so the regret does not grow with  $T$  at all, and the linear bound of Theorem 12 is driven purely by  $|\mathfrak{U}| = \infty$ . Two types break this, which is why the following result, though it concerns *full* information with  $K$  known, belongs here: it marks the limit of the naming account. The defender must now track which of two known opponents is currently more frequent, and no name settles that.

**Proposition 14** (two types already cost  $\sqrt{T}$ ). *Consider  $G_3$  with the two types  $\alpha^{(1)} = \alpha_{0, 1/2}$  and  $\alpha^{(2)} = \alpha_{1/2, 1}$ , so  $|\mathfrak{U}| = |K| = K_{\max} = 2$ , and let  $a_1, \dots, a_T$  be drawn i.i.d. uniformly from  $K$ . Then every online algorithm, randomized or deterministic, under either feedback model, and even one told  $K$  with its utility vectors at  $t = 0$ , satisfies*

$$\mathbb{E}[\text{Regret}(T)] \geq \frac{1}{2} \sqrt{\lceil T/2 \rceil} \quad (32)$$

*for every  $T \geq 2$ ; consequently some fixed sequence forces at least this much.*

*Proof.* The two intervals  $(0, \frac{1}{2}]$  and  $(\frac{1}{2}, 1]$  are disjoint, so by Lemma 10 every  $\mathbf{p}$  hits at most one type. Write  $N_i = |\{t : a_t = \alpha^{(i)}\}|$ , so  $N_1 + N_2 = T$ .

*Benchmark.* A fixed  $\mathbf{p}$  earns 0 when it hits and  $-1$  otherwise, so it earns at least  $-\min\{N_1, N_2\}$ , and this is attained:  $\mathbf{p} = (\frac{1}{4}, 0, \frac{3}{4})$  hits  $\alpha^{(1)}$  in all  $N_1$  of its rounds by Equation (25), and  $(\frac{3}{4}, 0, \frac{1}{4})$  hits  $\alpha^{(2)}$  in all  $N_2$  of its.

*Algorithm.* Condition on the history at round  $t$ ; it depends only on  $a_1, \dots, a_{t-1}$  and the algorithm's coins, so it is independent of  $a_t$ . The played  $\mathbf{p}_t$  hits at most one type and  $a_t$  is that type with probability  $\frac{1}{2}$ , so  $\Pr[\text{hit at } t] \leq \frac{1}{2}$  and the algorithm earns at most  $-T/2$  in expectation.

*Combining.* As  $\min\{N_1, N_2\} = \frac{1}{2}(T - |N_1 - N_2|)$ ,

$$\mathbb{E}[\text{Regret}(T)] \geq \frac{T}{2} - \mathbb{E}[\min\{N_1, N_2\}] = \frac{1}{2}\mathbb{E}[|N_1 - N_2|],$$

and  $N_1 - N_2$  is a simple random walk  $S_T$ , for which  $\mathbb{E}|S_T| \geq \sqrt{T/2}$  (Appendix A).  $\square$

The bound is tight in a strong sense:  $\Pr[\text{hit}] \leq \frac{1}{2}$  is an *equality* for any algorithm playing inside one of the two intervals, so all of them, follow-the-leader and one simply told  $K$  alike, have expected regret exactly  $\frac{1}{2}\mathbb{E}|N_1 - N_2| = \sqrt{T/(2\pi)}(1 + o(1))$ , the value Figure 4(b) measures.

So the transition at  $K_{\max} = 2$  is not about how hard a type is to identify, but about whether identification is the whole problem. The best upper bounds we can pair with it are  $O(\min\{T, T^{2/3}nM \log^{1/3}(nM)\})$ , from Balcan et al. [2, Theorem 6.1] with  $k = M$ , a finite universe being a known type set, and  $O(nK_{\max}\sqrt{T \log(nK_{\max})})$  from Theorem 6 under full information. Closing the gap to  $\Omega(\sqrt{T})$  is the finite- $M$  question of Section 10.

## 8 Block Revelation: Recovering Sublinear Regret

Theorem 12 shows that under partial information, the defender must receive some information about the identities of the attacker types. In particular, observing only the attacked target is not enough: the defender needs to know *which types are present*.

This section studies a small and realistic way of providing this information. Every  $B$  rounds, an oracle gives the defender the *set* of attacker types seen so far, together with their utility vectors. Importantly, the oracle does not reveal when a type attacked, how often it attacked, or whether it will attack again. Thus, the defender still cannot determine which type was responsible for any particular round.

This is important because identifying the attacker type from its observed best responses is exactly the difficulty considered by Balcan et al. [2, Section 6]. Our additional information does not remove this difficulty; it only tells the defender which types are present. As a result, the problem remains harder than the full-information setting, and the resulting regret rate is  $T^{2/3}$  rather than  $\sqrt{T}$ .

### 8.1 The model

**Problem 15** (Unknown- $K$  repeated SSG with block revelation). Fix a *revelation length*  $B \in \mathbb{N}$ , known to the defender,

and divide the horizon into  $N_B = \lceil T/B \rceil$  *revelation blocks*

$$\mathcal{B}_\nu = \{(\nu - 1)B + 1, \dots, \min\{\nu B, T\}\}, \quad \nu \in [N_B]. \quad (33)$$

Thus,  $\mathcal{B}_\nu$  is the  $\nu$ -th block of rounds, and a revelation occurs at the end of each block. The game is Problem 9 with the following additional feedback: at the end of  $\mathcal{B}_\nu$ , the defender receives

$$D_\nu = \{a_t : t \leq \min\{\nu B, T\}\} \subseteq K, \quad (34)$$

where  $a_t \in K$  denotes the attacker type that attacks in round  $t$ . Thus,  $D_\nu$  is the set of distinct attacker types that have appeared in the first  $\nu$  blocks. The defender also receives the utility vectors of all types  $\alpha \in D_\nu$ . We set  $D_0 = \emptyset$ . No other information is revealed. Within each round, the feedback remains partial.

Three remarks delimit the model. It is strictly weaker than Problem 4: even at  $B = 1$  the defender learns a round's type only when that type is new. Taking  $B \geq T$  makes the revelation useless and recovers Problem 9, so by Theorem 12 some bound on  $B$  is necessary. And the oracle is cumulative rather than incremental, but a defender that remembers  $D_{\nu-1}$  can convert between the two, so we use whichever is convenient.

### 8.2 The algorithm

Algorithm 2 keeps the outer skeleton of Algorithm 1: maintain a catalogue  $C$  of the attacker types the defender has been told about, run an inner regret-minimizer on  $\mathcal{E}(C; \epsilon)$ , and restart the inner learner whenever  $C$  grows. Two changes are forced by the model. First, the catalogue can enlarge only at the checkpoints  $t = B, 2B, \dots$ , so restarts happen at block boundaries rather than at the exact round a new type first attacks. Second, Polynomial Weights needs the round-level loss  $-U_d(b_{a_t}(\mathbf{p}), \mathbf{p})$ , which requires knowing  $a_t$ ; under partial information the defender does not see  $a_t$ , so this loss cannot be computed. We therefore replace Polynomial Weights by the partial-information learner  $\mathcal{A}(C)$  of Balcan et al. [2, Algorithm 1]. We fix  $\epsilon = 1/T$ : the approximation error  $2\epsilon L$  of Lemma 1 only shrinks with  $\epsilon$ , and Lemma 2 does not involve  $\epsilon$ .

We treat  $\mathcal{A}(C)$  as a black box. Its input is a catalogue  $C$  (which fixes the expert set  $\mathcal{E}(C; \epsilon)$ ); at each round it receives the observed target  $i_t$  and outputs the strategy  $\mathbf{p}_t \in \mathcal{E}(C; \epsilon)$  to play. Internally,  $\mathcal{A}(C)$  groups its rounds into estimation blocks and updates once per block, using each block's observations to form an unbiased estimate of the average per-strategy loss (via a barycentric-spanner probe on  $|C|$  random rounds of the block); that estimate is fed to a background full-information regret minimizer whose output drives the next block. The regret guarantee that this yields is the only property of  $\mathcal{A}$  the analysis uses, and we state and prove it inside the proof of Theorem 18, where it is applied.

### 8.3 Block structure

The analysis decomposes the horizon into two kinds of stretches: the blocks whose closing checkpoint names at least one new type, and the runs of blocks in between. Only the second kind carries useful learning; the first is charged to the additive term. Figure 1 illustrates the picture, and Definition 16 and Lemma 17 make it formal.

**Algorithm 2** BLOCK-GROW-CATALOGUE (partial information, revelation length  $B$ ).

**Require:** horizon  $T$ ; revelation length  $B$ ; game primitives; default strategy  $\mathbf{p}_{\text{def}} \in \mathcal{P}$

- 1:  $C \leftarrow \emptyset$ ;  $\epsilon \leftarrow 1/T$ ;  $\mathcal{A} \leftarrow \text{NULL}$   $\triangleright$  empty catalogue, no inner learner yet
- 2: **for**  $t = 1, \dots, T$  **do**
- 3:   **Play**  $\mathbf{p}_t \leftarrow \mathbf{p}_{\text{def}}$  if  $C = \emptyset$ , else the strategy  $\mathcal{A}$  prescribes for round  $t$
- 4:   **Observe** attacked target  $i_t = b_{a_t}(\mathbf{p}_t)$   $\triangleright$  identity of  $a_t$  not revealed
- 5:   pass  $i_t$  to  $\mathcal{A}$  as its round- $t$  feedback
- 6:   **if**  $t = \nu B$  for some  $\nu$ , or  $t = T$  **then**  $\triangleright$  revelation checkpoint
- 7:     receive  $D_\nu \subseteq K$  and the utilities of each  $\alpha \in D_\nu$
- 8:     **if**  $D_\nu \neq C$  **then**  $\triangleright$  new type(s) appeared during  $\mathcal{B}_\nu$
- 9:        $C \leftarrow D_\nu$
- 10:       $\mathcal{A} \leftarrow \mathcal{A}(C)$   $\triangleright$  restart on enlarged catalogue; state built during  $\mathcal{B}_\nu$  is discarded
- 11:    **end if**
- 12: **end if**
- 13: **end for**

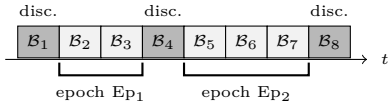


Figure 1: The block decomposition. Shaded blocks are the *discovery blocks*, their closing checkpoint enlarges the catalogue, and unshaded blocks are *clean*. The maximal run of clean blocks strictly between two consecutive discovery blocks forms an *epoch*; inside an epoch the catalogue is fixed and Algorithm 2 runs one uninterrupted inner instance  $\mathcal{A}(C_j)$ .

**Definition 16** (discovery blocks and epochs). Call  $\mathcal{B}_\nu$  a *discovery block* if  $D_\nu \supsetneq D_{\nu-1}$  and a *clean block* otherwise. Let  $\beta_1 < \dots < \beta_m$  index the discovery blocks. The  $j$ -th *epoch*  $\text{Ep}_j$  is the union of the clean blocks strictly between the  $j$ -th and  $(j+1)$ -th discovery block,

$$\text{Ep}_j = \bigcup_{\nu=\beta_j+1}^{\beta_{j+1}-1} \mathcal{B}_\nu, \quad L_j := |\text{Ep}_j|, \quad (35)$$

with  $\beta_{m+1} := N_B + 1$ . Write  $C_j := D_{\beta_j}$  for the catalogue Algorithm 2 carries through  $\text{Ep}_j$ .

**Lemma 17** (decomposition). *Let  $K' = |\{a_1, \dots, a_T\}| \leq K_{\max}$  be the number of types that actually appear in  $\mathbf{a}$ . For every oblivious  $\mathbf{a} \in K^T$ :*

- (a) *Partition.  $\beta_1 = 1$ ,  $m \leq K'$ , and  $[1, T] = (\bigsqcup_j \mathcal{B}_{\beta_j}) \sqcup (\bigsqcup_j \text{Ep}_j)$  with  $\sum_j L_j \leq T$  and the discovery blocks contributing at most  $mB \leq K'B$  rounds in total.*
- (b) *Clean epochs. Throughout  $\text{Ep}_j$  the working catalogue equals  $C_j$ ; every attacker that appears in  $\text{Ep}_j$  belongs to  $C_j$ ; and Algorithm 2 runs one uninterrupted inner instance  $\mathcal{A}(C_j)$  that was started at the first round of  $\text{Ep}_j$  and every observation of which at a round of  $\text{Ep}_j$  comes from a type inside  $C_j$ .*
- (c) *Obliviousness. The discovery blocks and the epochs are functions of  $\mathbf{a}$  alone, independent of the defender's plays and randomization.*

*Proof.* (a)  $D_0 = \emptyset \neq D_1$  makes  $\mathcal{B}_1$  a discovery block; each discovery block adds at least one type, so  $m \leq K'$ . Every block is either a discovery block or lies in exactly one epoch, and holds  $\leq B$  rounds.

(b) If  $\mathcal{B}_\nu$  is clean then  $D_\nu = D_{\nu-1}$ , so any  $t \in \mathcal{B}_\nu$  has  $a_t \in D_{\nu-1}$ ; no new type surfaces inside a clean block. Algorithm 2 changes  $C$  only at discovery blocks, so  $C = C_j$  throughout  $\text{Ep}_j$ , and by the same argument  $a_t \in C_j$  for every  $t \in \text{Ep}_j$ . The instance is restarted at the checkpoint closing  $\beta_j$  and not again inside  $\text{Ep}_j$ . The restriction “at a round of  $\text{Ep}_j$ ” matters because the instance keeps running into the following discovery block, where contaminated observations can enter its state, but Algorithm 2 discards it at the next checkpoint, so no contamination survives into a later epoch or enters the regret bound.

(c) By Equation (34), each  $D_\nu$  is a function of  $a_1, \dots, a_{\min\{\nu B, T\}}$ ; the same holds for the predicate “ $\mathcal{B}_\nu$  is a discovery block” and for the  $\beta_j$  and epochs.  $\square$

## 8.4 Main result

**Theorem 18** (block revelation restores no-regret). *Let the defender play Algorithm 2 in Problem 15. For any oblivious  $\mathbf{a} \in K^T$  with  $|K| \leq K_{\max}$ , and  $K' \leq K_{\max}$  the number of distinct types appearing,*

$$\mathbb{E}[\text{Regret}(T)] \leq cn K'^{4/3} T^{2/3} \log^{1/3}(nK') + 2K'B \quad (36)$$

for an absolute constant  $c$ ; in particular  $\mathbb{E}[\text{Regret}(T)] = O(nK_{\max}^{4/3} T^{2/3} \log^{1/3}(nK_{\max}) + K_{\max}B)$ .

*Proof.* Fix an oblivious adversarial sequence  $\mathbf{a} \in K^T$  with  $|K| \leq K_{\max}$  and let  $K'$  be the number of distinct types that appear in  $\mathbf{a}$ . Adopt the notation of Definition 16 and Lemma 17. For any set of rounds  $S \subseteq [1, T]$  write

$$\text{Regret}(S) = \sum_{t \in S} U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*) - \mathbb{E} \left[ \sum_{t \in S} U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t) \right], \quad (37)$$

so that  $\text{Regret}([1, T]) = \text{Regret}(T)$  and Regret is additive over disjoint unions. The proof proceeds in ten steps.

**Step 1 (Block decomposition).** By Lemma 17, the horizon splits as the disjoint union of the  $m \leq K'$  discovery blocks  $\mathcal{B}_{\beta_j}$  and the  $m$  epochs  $\text{Ep}_j$ . Additivity of Equation (37) therefore gives

$$\text{Regret}(T) = \sum_{j=1}^m \text{Regret}(\text{Ep}_j) + \sum_{j=1}^m \text{Regret}(\mathcal{B}_{\beta_j}). \quad (38)$$

Steps 2–5 control the first sum; Step 6 controls the second.

**Step 2 (Catalogue is constant and correct inside an epoch).** By Lemma 17, throughout  $\text{Ep}_j$  the working catalogue equals  $C_j := D_{\beta_j}$ , every attacker  $a_t$  with  $t \in \text{Ep}_j$  lies in  $C_j$ , and a single instance  $\mathcal{A}(C_j)$  started at the first round of  $\text{Ep}_j$  runs uninterrupted through the epoch on observations drawn from  $C_j$ . Also by Lemma 17, the epoch and its indexing are functions of  $\mathbf{a}$  alone, so the sub-sequence  $(a_t)_{t \in \text{Ep}_j}$  is oblivious.

**Step 3 (Regret guarantee for the inner learner).** The only property of  $\mathcal{A}(C)$  that the analysis uses is an oblivious no-regret bound on the fixed-catalogue subproblem, the partial-information counterpart of Proposition 3. Importing it from Balcan et al. [2, Theorem 6.1] needs one adjustment: their algorithm is stated for a horizon  $H$  known in advance, because it splits  $[1, H]$  into a horizon-tuned number of *estimation blocks* (each spending  $|C|$  rounds probing the barycentric spanner and the rest playing what a background full-information sub-routine proposes), with the block count chosen to balance exploration and exploitation at that particular  $H$ . The epoch  $\text{Ep}_j$ , by contrast, closes only when a new type appears, so its length  $L_j$  is unknown to  $\mathcal{A}(C_j)$  when it starts. We need a bound holding *simultaneously* at every prefix  $L \leq T$ , with  $L$  on the right-hand side.

**Lemma 19** (anytime partial-information learner). *For every finite catalogue  $C$  and every  $\epsilon \leq 1/T$  there is a variant of  $\mathcal{A}(C)$  that does not require its horizon in advance, and an absolute constant  $c_4$ , such that on any oblivious sequence  $a'_1, a'_2, \dots$  with  $a'_t \in C$  and for every  $L \leq T$  simultaneously,*

$$\begin{aligned} \max_{\mathbf{p} \in \mathcal{P}} \sum_{t=1}^L U_d(b_{a'_t}(\mathbf{p}), \mathbf{p}) - \mathbb{E} \left[ \sum_{t=1}^L U_d(b_{a'_t}(\mathbf{p}_t), \mathbf{p}_t) \right] \\ \leq c_4 L^{2/3} n |C| \log^{1/3}(n|C|). \end{aligned} \quad (39)$$

The benchmark on the left is the best fixed strategy in hindsight over  $\mathcal{P}$  on the sequence  $a'_1, \dots, a'_L$ , i.e. the epoch-local optimum; passing to the global  $\mathbf{p}^*$  of the theorem is Step 4b.

*Proof of Lemma 19.* Two ingredients suffice: (i) Balcan's bound, though stated at the end of a run tuned to a horizon  $H$ , is actually valid at every intermediate prefix of that run; (ii) a doubling wrapper removes the tuning.

(i) *Balcan's bound holds at every prefix.* Fix  $H$  and run  $\mathcal{A}$  tuned to it. The core of Balcan et al. [2, Lemma 6.2]'s argument is a chain of inequalities which, after any  $\tau_0$  completed estimation blocks, bounds the algorithm's expected loss so far by the loss of the best fixed strategy over those same blocks plus two error terms: one that counts the  $|C|$  exploration rounds consumed per block, and one that carries the regret of the background full-information sub-routine after  $\tau_0$  estimated loss vectors. Both grow monotonically in  $\tau_0$ , and by Proposition 3 the sub-routine is itself anytime, so its regret bound holds at every  $\tau_0$ , not only at Balcan's fully-tuned choice. Reading the chain at any  $\tau_0$  therefore reproduces their bound with  $L$  in place of  $H$ ; a possibly-incomplete last block and the discretization error  $2\epsilon L \leq 2$  from Lemma 1 contribute a constant that is absorbed. What fixes the exponent is the horizon-tuned choice of block count: there is an absolute constant  $c_0$  such that a run tuned to  $H$  satisfies

$$\text{regret at prefix } L \leq c_0 H^{2/3} n |C| \log^{1/3}(n|C|) \quad (40)$$

for every  $L \leq H$ . The right-hand side depends on  $H$ , not  $L$ : at short prefixes we are still paying for the horizon we tuned to.

(ii) *Doubling wrapper.* To replace  $H^{2/3}$  by  $L^{2/3}$ , run  $\mathcal{A}$  in stages of exponentially growing length: stage  $s = 0, 1, 2, \dots$  has length  $2^s$  and is a fresh copy of Balcan's algorithm tuned to horizon  $2^s$ ; at the end of stage  $s$  we discard its state and

start stage  $s+1$ . Fix any  $L$  and let  $S$  be the stage containing round  $L$ ; stages  $0, \dots, S-1$  occupy  $2^S - 1$  rounds, all  $\leq L$ , so  $2^S \leq L$ .

Regret is subadditive across stages: for any  $\mathbf{p} \in \mathcal{P}$  the payoff over  $[1, L]$  decomposes as a sum of per-stage payoffs, and  $\max_{\mathbf{p} \in \mathcal{P}}$  of a sum is at most the sum of per-stage  $\max_{\mathbf{p} \in \mathcal{P}}$ 's. Subtracting the algorithm's payoff stage by stage,

$$\text{regret over } [1, L] \leq \sum_{s=0}^S \text{regret of stage } s \leq c_0 \phi \sum_{s=0}^S 2^{2s/3},$$

where  $\phi := n|C| \log^{1/3}(n|C|)$  and each stage's regret comes from Equation (40) with  $H = 2^s$ . The remaining sum is a geometric series with ratio  $2^{2/3} > 1$ , so

$$\sum_{s=0}^S 2^{2s/3} \leq \frac{2^{2/3}}{2^{2/3} - 1} \cdot 2^{2S/3} \leq \frac{2^{2/3}}{2^{2/3} - 1} \cdot L^{2/3},$$

the last step using  $2^S \leq L$ . Setting  $c_4 := 2^{2/3} c_0 / (2^{2/3} - 1) < 2.71 c_0$  yields Equation (39).  $\square$

The doubling wrapper costs a constant factor of  $\approx 2.71$ ; the exponent  $2/3$  and every other factor in Equation (39) are preserved.

**Step 4a (Applying Lemma 19 to an epoch).** Fix  $j$ . By Step 2 the sub-sequence  $(a_t)_{t \in \text{Ep}_j}$  is oblivious and takes values in  $C_j$ , so the hypotheses of Lemma 19 are met with catalogue  $C_j$ , prefix length  $L_j$ , and the sub-sequence in the role of  $a'_1, \dots, a'_{L_j}$ . It yields

$$\begin{aligned} \max_{\mathbf{p} \in \mathcal{P}} \underbrace{\sum_{t \in \text{Ep}_j} U_d(b_{a_t}(\mathbf{p}), \mathbf{p}) - \mathbb{E} \left[ \sum_{t \in \text{Ep}_j} U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t) \right]}_{=:\text{OPT}_j} \\ \leq c_4 L_j^{2/3} n |C_j| \log^{1/3}(n|C_j|). \end{aligned} \quad (41)$$

**Step 4b (Passing from  $\text{OPT}_j$  to  $\mathbf{p}^*$ ).** The bound Equation (41) is against  $\text{OPT}_j$ , the *epoch-local* maximum: the best cumulative utility on  $\text{Ep}_j$  over *any*  $\mathbf{p} \in \mathcal{P}$ . The regret of Equation (37), on the other hand, is measured against the global benchmark  $\mathbf{p}^*$  defined in Equation (3), which is optimal over the full horizon  $[1, T]$ , not over  $\text{Ep}_j$ . The two disagree in general:  $\text{OPT}_j \neq \sum_{t \in \text{Ep}_j} U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*)$ , because a strategy tuned to the specific mix of types active in  $\text{Ep}_j$  can beat  $\mathbf{p}^*$  on that epoch even though  $\mathbf{p}^*$  dominates it over  $[1, T]$ .

The passage from the local  $\text{OPT}_j$  to the global  $\mathbf{p}^*$  is however immediate. Since  $\mathbf{p}^* \in \mathcal{P}$ , it is a feasible competitor in the maximum defining  $\text{OPT}_j$ , so

$$\text{OPT}_j \geq \sum_{t \in \text{Ep}_j} U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*). \quad (42)$$

The local-optimum bound of Equation (41) therefore implies a bound against the global benchmark, since  $\mathbf{p}^*$  lives in the same feasible set  $\mathcal{P}$  as the maximizer of  $\text{OPT}_j$ .

**Step 5 (Per-epoch combined bound).** Combining Equations (41) and (42) with the definition Equation (37) gives, for every  $j \in [m]$ ,

$$\text{Regret}(\text{Ep}_j) \leq c_4 L_j^{2/3} n |C_j| \log^{1/3}(n|C_j|). \quad (43)$$

**Step 6 (Discovery blocks).** Since  $U_d(i, \mathbf{p}) \in [-1, 1]$  for every target  $i$  and every  $\mathbf{p} \in \mathcal{P}$ , each round contributes at most 2 to Equation (37). A discovery block holds at most  $B$  rounds, and by Lemma 17 there are  $m \leq K'$  discovery blocks, so

$$\sum_{j=1}^m \text{Regret}(\mathcal{B}_{\beta_j}) \leq 2mB \leq 2K'B. \quad (44)$$

This is the price of batching: no learning is attempted inside a discovery block, and Theorem 22 shows that no algorithm can do better.

**Step 7 (Uniform bound on the catalogue size).** Since the catalogue only grows and the terminal catalogue  $D_{\beta_m}$  contains at most  $K'$  types,  $|C_j| \leq K'$  for every  $j$ . Hence

$$|C_j| \log^{1/3}(n|C_j|) \leq K' \log^{1/3}(nK') \quad \text{uniformly in } j, \quad (45)$$

which lets the catalogue-dependent factor be pulled out of the sum in the next step. The size bound on  $\mathcal{E}(C_j; \epsilon)$  that would otherwise appear here is already folded into the statement of Lemma 19.

**Step 8 (Summing across epochs).** Substituting Equations (43) to (45) into Equation (38),

$$\mathbb{E}[\text{Regret}(T)] \leq c_4 n K' \log^{1/3}(nK') \sum_{j=1}^m L_j^{2/3} + 2K'B. \quad (46)$$

**Step 9 (Hölder's inequality).** The remaining sum  $\sum_j L_j^{2/3}$  is controlled by Hölder's inequality, which we recall. For real numbers  $x_1, \dots, x_m, y_1, \dots, y_m$  and conjugate exponents  $p, q > 1$  with  $1/p + 1/q = 1$ ,

$$\sum_{j=1}^m |x_j y_j| \leq \left( \sum_{j=1}^m |x_j|^p \right)^{1/p} \left( \sum_{j=1}^m |y_j|^q \right)^{1/q}. \quad (47)$$

Set  $x_j = 1$  and  $y_j = L_j^{2/3}$ , and take  $p = 3$ ,  $q = 3/2$ ; then  $\sum |x_j|^p = m$  and  $\sum |y_j|^q = \sum L_j$ , so

$$\sum_{j=1}^m L_j^{2/3} \leq m^{1/3} \left( \sum_{j=1}^m L_j \right)^{2/3} \leq K'^{1/3} T^{2/3}, \quad (48)$$

where the last step uses  $m \leq K'$  and  $\sum_j L_j \leq T$  from Lemma 17. Equality in Equation (48) holds when all epochs have equal length, and this is where the multiplicative  $K'^{1/3}$  factor of the theorem is generated.

**Step 10 (Final substitution).** Substituting Equation (48) into Equation (46) gives

$$\mathbb{E}[\text{Regret}(T)] \leq c_4 n K'^{4/3} T^{2/3} \log^{1/3}(nK') + 2K'B,$$

which is Equation (36) with  $c = c_4$ . Using  $K' \leq K_{\max}$  gives the  $K_{\max}$ -form of the theorem.  $\square$

*Remark 20* (reading the bound). Equation (36) differs from Balcan et al. [2, Theorem 6.1] at  $k = K_{\max}$  in two places. The multiplicative factor  $K_{\max}^{1/3}$  is the price of learning  $K$  incrementally, produced by Hölder over the epochs (analogous

to the  $\sqrt{K_{\max}}$  that Cauchy–Schwarz produces in Theorem 6). The additive  $2K_{\max}B$  is the price of batching, tight up to a factor  $2B/(B-2)$  by Theorem 22. The exponent  $2/3$  is inherited from the inner learner and untouched by the unknown catalogue.

Two consequences. First, taking  $B = 1$  does *not* recover Theorem 6: even then the defender still cannot attribute a round to a specific known type, so the inner learner remains the frequency-estimating one, which fixes the  $2/3$  exponent regardless of the decomposition. Second, Algorithm 2 never reads  $K_{\max}$ : the bound is stated in  $K'$  and is instance-adaptive in the sense of Corollary 8, so the defender needs only  $T$  and  $B$ .

## 8.5 How coarse may the revelation be?

Equation (36) has two pieces: a learning term of order  $nK_{\max}^{4/3} T^{2/3} \log^{1/3}(nK_{\max})$  and a batching term  $2K_{\max}B$ . The natural question is how large  $B$  can be made before the second overtakes the first.

**Corollary 21** (free batching, and the number of oracle calls). *The batching term of Equation (36) is dominated as soon as  $B \leq B^*$ , where*

$$\begin{aligned} B^* &= \frac{1}{2} c n K_{\max}^{1/3} T^{2/3} \log^{1/3}(nK_{\max}) \\ &= \Theta(nK_{\max}^{1/3} T^{2/3} \log^{1/3}(nK_{\max})), \end{aligned} \quad (49)$$

and then  $\mathbb{E}[\text{Regret}(T)] = O(nK_{\max}^{4/3} T^{2/3} \log^{1/3}(nK_{\max}))$ : batching costs nothing at this rate. Equivalently, the defender need consult the oracle only

$$N_B = \lceil T/B^* \rceil = O\left( \frac{T^{1/3}}{n K_{\max}^{1/3} \log^{1/3}(nK_{\max})} \right) \quad (50)$$

times. Finally, Theorem 18 is sublinear in  $T$  iff  $nK_{\max}^{4/3} \log^{1/3}(nK_{\max}) = o(T^{1/3})$  and  $K_{\max}B = o(T)$ ; for  $K_{\max} = O(1)$  and  $B \leq B^*$  the rate is  $\tilde{O}(nT^{2/3})$ .

*Proof.* Solving  $2K_{\max}B \leq c n K_{\max}^{4/3} T^{2/3} \log^{1/3}(nK_{\max})$  for  $B$  gives Equation (49), and Equation (50) is  $\lceil T/B^* \rceil$ . The sublinearity criterion is Equation (36) at  $K' = K_{\max}$ , divided by  $T$ , requiring each term to vanish.  $\square$

## 8.6 The additive term is necessary

The  $\Theta(K_{\max}B)$  term is not an artefact of the analysis but an inherent cost of the model. The construction that certifies this reuses  $G_3$ : each new type's threshold interval is nested strictly inside the previous one's, so a single coverage vector, the midpoint of the innermost interval, is optimal against all  $K_{\max}$  types simultaneously, making the benchmark's regret zero. Every type must nevertheless be located from scratch inside its block, since the oracle reveals only that a type exists and provides its utility vector, not the interval it prefers.

**Theorem 22** (a matching lower bound in  $B$ ). *Let  $G_3$  and  $\mathcal{U}$  be as in Theorem 12. For every  $B \geq 1$ , every  $K_{\max} \geq 1$  and every  $T \geq K_{\max}B$ , there is a distribution over instances of Problem 15 with  $|K| = K_{\max}$  on which every online algorithm suffers*

$$\mathbb{E}[\text{Regret}(T)] \geq K_{\max} (B - 2). \quad (51)$$

*Proof.* Put  $M = 2^B$ , draw  $k_1, \dots, k_{K_{\max}}$  independently and uniformly from  $[M]$ , and define nested intervals  $J_0 = (0, 1] \supset J_1 \supset \dots$  by letting  $J_j$  be the  $k_j$ -th of the  $M$  equal sub-intervals of  $J_{j-1}$ :

$$J_j = \left( \ell_{j-1} + \frac{k_j - 1}{M} |J_{j-1}|, \ell_{j-1} + \frac{k_j}{M} |J_{j-1}| \right], \quad (52)$$

$\ell_{j-1}$  and  $|J_{j-1}|$  being the left endpoint and length of  $J_{j-1}$ . Let  $\alpha^{[j]}$  be the threshold type with interval  $J_j$ , put  $K = \{\alpha^{[1]}, \dots, \alpha^{[K_{\max}]}\}$ , and let  $a_t = \alpha^{[j]}$  for  $t \in \mathcal{B}_j$  and  $a_t = \alpha^{[K_{\max}]}$  for  $t > K_{\max}B$ . The sequence is fixed once the  $k_j$  are drawn, so the adversary is oblivious, and each  $J_j$  is a *proper* sub-interval of  $J_{j-1}$ , so the  $K_{\max}$  types are distinct. As before we bound deterministic algorithms and appeal to Yao.

*The benchmark is 0.* Let  $\mu$  be the midpoint of the innermost interval  $J_{K_{\max}}$  and  $\mathbf{p}^* = (\mu, 0, 1 - \mu)$ . Nesting gives  $r_1^{[j]} \leq r_1^{[K_{\max}]} < \mu$  and  $1 - r_2^{[j]} \leq 1 - r_2^{[K_{\max}]} < 1 - \mu$  for every  $j$ , so  $\mathbf{p}^*$  hits every type by Equation (25), under any tie-breaking rule, and regret again counts misses.

*Each block is a fresh search.* When  $\mathcal{B}_j$  begins, the checkpoint closing  $\mathcal{B}_{j-1}$  has revealed  $\alpha^{[1]}, \dots, \alpha^{[j-1]}$ , hence  $J_{j-1}$ , and nothing about  $\alpha^{[j]}$ , which has not yet attacked. Moreover the whole transcript of  $\mathcal{B}_1, \dots, \mathcal{B}_{j-1}$  is a function of  $k_1, \dots, k_{j-1}$  alone: inductively, each play is determined by earlier observations, and each observation by that play and the interval then attacking. As  $k_j$  is drawn independently, it remains uniform on  $[M]$  conditionally on everything the defender has seen and been told.

Condition on that transcript. Within  $\mathcal{B}_j$  the defender's plays are deterministic functions of the observations made inside  $\mathcal{B}_j$ , the only attacker is  $\alpha^{[j]}$ , and its interval is one of  $M$  disjoint sub-intervals of  $J_{j-1}$ , so Lemma 10 gives Equation (27). The conditioning has absorbed everything the oracle said before  $\mathcal{B}_j$ , and it says nothing during  $\mathcal{B}_j$ , so within the block the interaction is a plain protocol of the kind Lemma 11 covers. Lemma 11 with  $M = 2^B$  and horizon  $B$  gives  $\mathbb{E}[\#\text{misses in } \mathcal{B}_j] \geq B - (2^{B+1} - B - 2)/2^B > B - 2$ . Summing over  $j \in [K_{\max}]$  and averaging over the algorithm's randomness gives Equation (51).  $\square$

*Remark 23* (the additive term is tight). Theorems 18 and 22 bracket the cost of batching between  $K_{\max}(B-2)$  and  $2K_{\max}B$ , so it is  $\Theta(K_{\max}B)$  for  $B \geq 3$ , the two differing by  $2B/(B-2)$ , 4 at  $B = 4$  and  $2 + o(1)$  thereafter; for  $B \leq 2$  the lower bound is vacuous. The two endpoints are worth naming. At  $K_{\max} = 1$ ,  $B = T$  the bound reads  $\mathbb{E}[\text{Regret}(T)] \geq T - 2$ , so Theorem 12 is exactly the  $B \rightarrow T$  limit of Theorem 22. At  $B = 1$  the additive term is  $2K_{\max}$  and Equation (36) becomes  $O(nK_{\max}^{4/3}T^{2/3} \log^{1/3}(nK_{\max}) + K_{\max})$ , the partial-information counterpart of Theorem 6, and to our knowledge the first sublinear bound for an unknown attacker set under partial-information feedback.

## 9 Experiments

We report three groups of experiments. Sections 9.2 and 9.3 test Algorithm 1 and Algorithm 2 against the envelopes of Theorems 6 and 18, first at  $n = 3$  and then, with the oracle implementation of Appendix B, at  $n = 10$  and  $n = 20$ .

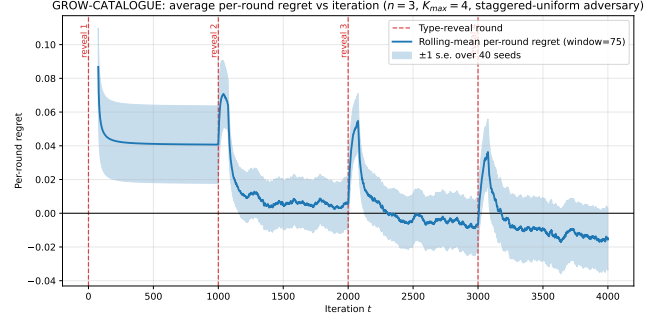


Figure 2: Rolling-window per-round regret of GROW-CATALOGUE under the staggered-uniform adversary ( $n = 3$ ,  $K_{\max} = 4$ ,  $T = 4000$ , average over 40 seeds, window = 75). Red dashed lines mark the discovery rounds 0, 1000, 2000, 3000. Each reveal causes a visible transient spike, followed by PW re-convergence inside the new epoch.

Section 9.4 instead plays the gadget  $G_3$  of Section 7.2 exactly, checking that the lower bounds of Theorems 12 and 22 and Proposition 14 are attained rather than artefacts of the analysis.

### 9.1 Setup

We implement Algorithm 1 in Python, along with two reference algorithms: the known- $K$  oracle of Balcan et al. [2, Sec. 5], which receives the full set  $K$  at  $t = 1$ ; and a *uniform* baseline that plays  $\mathbf{p}_{\text{def}} = (R/n, \dots, R/n)$  every round. Unless stated otherwise, we use  $n = 3$  targets,  $R = 1$  resource, and attacker utilities drawn i.i.d. from  $\text{Unif}[-1, 1]$  with  $u_\alpha^u(i) > 0 > u_\alpha^c(i)$  to ensure covering is costly for the attacker. The extreme-point set  $\mathcal{E}(C; \epsilon)$  is computed by direct enumeration of intersections of  $(n-1)$ -subsets of the defining hyperplanes, which is tractable at  $n = 3$ .

We use a *staggered-uniform* adversary: the horizon  $[0, T]$  is partitioned into  $K_{\max}$  phases of length  $T/K_{\max}$  each, and in phase  $i$  the attacker is drawn uniformly at random from the first  $i$  types  $\{\alpha_1, \dots, \alpha_i\}$ . Thus  $\alpha_1$  is present in every phase,  $\alpha_2$  from phase 2 onward, and so on. This design produces a clean sequence of discovery rounds at  $t = 0, T/K_{\max}, \dots, (K_{\max} - 1)T/K_{\max}$ , while still mixing old and new types after each reveal so that the benchmark  $\mathbf{p}^*$  is a genuine compromise rather than a specialist strategy.

### 9.2 Full information at $n = 3$

Figure 2 shows that the algorithm's per-round regret tracks the epoch structure of the proof: each discovery round produces a bounded transient, and the PW sub-routine re-converges within the new epoch. The spikes shrink with  $i$  because, by epoch  $i$ , the algorithm already has a good PW distribution on most of the new (refined) extreme points.

Figure 3 plots cumulative  $\text{Regret}(T)$  versus  $T$  with the  $C\sqrt{T}$  envelope from Theorem 6 overlaid. The empirical regret stays visibly below the envelope for all  $T$  tested, and the ratio  $\text{Regret}(T)/\sqrt{T}$  remains bounded (in fact, it decreases, indicating that the stochastic staggered-uniform adversary

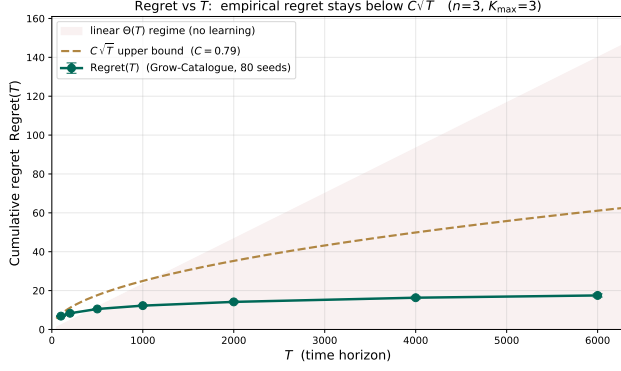


Figure 3: Cumulative regret versus horizon  $T$  for GROW-CATALOGUE ( $n = 3$ ,  $K_{\max} = 3$ , uniform random adversary, 80 seeds). The green curve stays strictly below the  $C\sqrt{T}$  upper envelope implied by Theorem 6, which is in turn well below the linear-regret regime that would result without learning.

is far from the worst case). This is entirely consistent with Theorem 6, which is our upper bound

Table 1 compares the four algorithms at  $T = 4000$ . The regret of GROW-CATALOGUE is within roughly 15%–40% of the known- $K$  oracle and an order of magnitude better than the uniform baseline. The two GROW-CATALOGUE variants produce regret that differs only by Monte-Carlo noise, as predicted by the observational equivalence of the two recomputation schemes.

| Algorithm                              | $\mathbb{E}[\text{Regret}(T)]$ | Ratio |
|----------------------------------------|--------------------------------|-------|
| Known- $K$ oracle of Balcan et al. [2] | $14.51 \pm 5.00$               | 1.00  |
| GROW-CAT. (total)                      | $16.57 \pm 5.40$               | 1.14  |
| Uniform baseline                       | $292.77 \pm 350.28$            | 20.18 |

Table 1: Cumulative regret at  $T = 4000$ ,  $n = 3$ ,  $K_{\max} = 3$ , uniform-random adversary, mean  $\pm$  std over 15 seeds. The known- $K$  oracle receives  $K$  in advance; GROW-CATALOGUE attains comparable regret *without* this information.

### 9.3 Scaling to $n = 10$ and $n = 20$

The runs of Section 9.2 were at  $n = 3$  targets, limited not by the online algorithm but by the expert set: Algorithm 1 needs  $\mathcal{E}(C; \epsilon)$  explicitly, and building it by enumerating  $(n-1)$ -subsets of the defining hyperplanes costs  $7.3 \times 10^{14}$  linear solves at  $n = 10$ . Neither algorithm actually needs that set, however, each only needs to *optimize a linear function over it*, which one call to a Bayesian-Stackelberg solver does. Appendix B gives the construction and Appendix C the resulting runs at  $n = 10$  and  $n = 20$ ; here we report what they show and then test the lower bounds directly.

At  $n = 10$  and  $n = 20$  with  $K_{\max} = 4$ , Algorithm 1 stays below the  $C\sqrt{t}$  envelope of Theorem 6 throughout, and Algorithm 2 below the  $Ct^{2/3}$  envelope of Theorem 18. Fitting the growth over  $T \in \{250, \dots, 2000\}$  gives exponent 0.664 for block revelation, the  $2/3$  of Theorem 18, to the resolution of the experiment, and 0.317 for full information, comfortably inside the  $1/2$  of Theorem 6. Sweeping  $B$  at fixed  $T$  gives

regret growing linearly in  $B$  with slope 0.47, well below the worst-case  $2K' = 8$ , which is expected: the worst case is attained only by the nested adversary of Theorem 22, exhibited separately in Figure 4(c).

### 9.4 The lower bounds, in simulation

Figure 4 plays the gadget  $G_3$  of Section 7.2 for real: attacker types are built from the utility vectors of Equation (22), best responses are computed from those utilities, and the defender’s payoff is read off the attacked target. The defender is the optimal one for this game, namely the midpoint version-space search of Remark 13, and the expectation over the hidden type is computed exactly, by averaging over all  $M$  candidates rather than by sampling.

Panel (a) is Theorem 12: with  $M = 2^T$  the measured expected regret equals  $T - 2 + (T + 2)2^{-T}$  for every  $T \leq 14$ , to machine precision, confirming both the lower bound and the claim in Remark 13 that it is exactly attained.

Panel (b) is Proposition 14, and it is the one panel that shows a  $T$ -dependence rather than a saturation. With  $|K| = 2$  on the two-element universe, follow the leader has measured regret 1.57, 6.35 and 25.60 at  $T = 16$ , 256 and 4096, against the exact optimum  $\frac{1}{2}\mathbb{E}|N_1 - N_2| = 1.571, 6.377, 25.531$  and the floor  $\frac{1}{2}\sqrt{\lfloor T/2 \rfloor} = 1.41, 5.66, 22.63$ . The dotted line is the same universe with  $|K| = 1$ , where the regret is 0.5 at every horizon. The two curves are the phase transition of Section 7.5.

Panel (c) is Theorem 22: on the nested-interval instance the measured regret tracks  $|K|(B - 2)$  across  $|K| \in \{1, 2, 3, 4\}$  and  $B \in \{4, 8, 16, 32\}$ , so the additive  $\Theta(K_{\max}B)$  of Theorem 18 is real and not an artefact of the analysis.

## 10 Discussion and Open Problems

We studied the repeated Stackelberg security game in a new, open regime in which the set of attacker types is unknown to the defender a priori but known to have bounded cardinality. We presented GROW-CATALOGUE, a simple online algorithm that incrementally discovers attacker types and re-partitions the coverage polytope accordingly, and we gave a full proof that its expected regret under full-information feedback is at most  $O(nK_{\max}\sqrt{T\log(nK_{\max})})$ . This is exactly the known- $K$  bound of Balcan et al. [2, Theorem 5.1] with the known cardinality  $k$  replaced by our known upper bound  $K_{\max}$ , inflated by the  $\sqrt{K_{\max}}$  that the epoch decomposition costs. The bound is sublinear whenever  $n^2K_{\max}^2\log(nK_{\max}) = o(T)$ , and the matching  $\Omega(T)$  lower bound of Balcan et al. [2, Theorem 7.1] rules out further improvements on the unbounded side. Our experiments confirm the analytical scaling: the empirical regret stays below the  $C\sqrt{T}$  envelope and tracks the epoch structure of the proof.

Beyond that full-information picture, Sections 7 and 8 locate the boundary of learnability for the unknown- $K$  repeated security game. On one side, Theorem 6: an unknown attacker set of bounded size costs only  $\sqrt{K_{\max}}$ , provided the defender is told after each round which type attacked. On the other, Theorem 12: told nothing but the attacked target, every algorithm loses all but two of the  $T$  rounds, not because the adversary has many types, but because the defender cannot

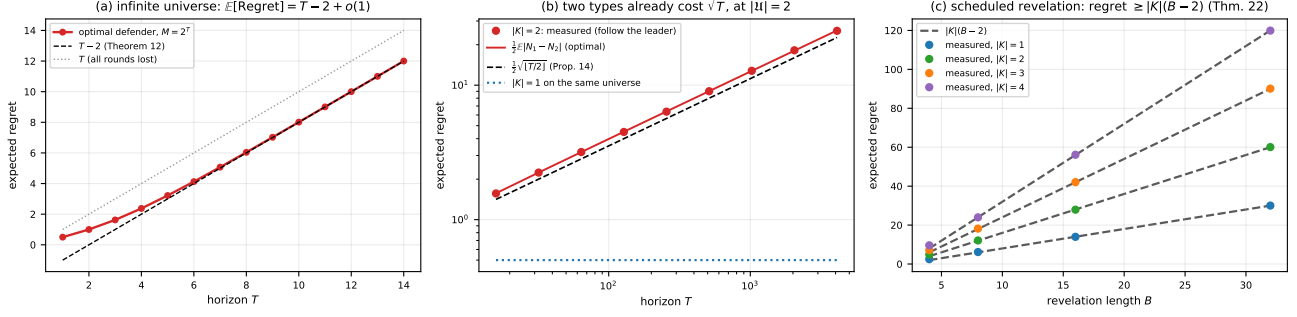


Figure 4: The gadget  $G_3$  in simulation, against the optimal defender. (a) Theorem 12: with  $|\mathcal{U}_M| = 2^T$  the regret is  $T - 2 + o(1)$ . (b) Proposition 14: two types on a two-element universe cost  $\Theta(\sqrt{T})$ , one type costs 0.5 at every horizon. (c) Theorem 22: on the nested instance the regret matches  $|K|(B - 2)$  (dashed lines). Panel (b) has a logarithmic vertical axis.

| Feedback | Set $K$                                        | Regret                                          |
|----------|------------------------------------------------|-------------------------------------------------|
| full     | known, $ K  = k$                               | $O(\sqrt{Tn^2k \log nk})$                       |
| partial  | known, $ K  = k$                               | $O(T^{2/3}nk \log^{1/3}nk)$                     |
| full     | known, $ K  = 2^{T+1} - 2$                     | $\Omega(T)$                                     |
| full     | unknown, $\leq K_{\max}$                       | $O(nK_{\max} \sqrt{T \log nK_{\max}})$          |
| partial  | unknown, $\leq K_{\max}$                       | $\Omega(T)$ , even $K_{\max}=1$                 |
| full     | known, $ K = \mathcal{U} =2$                   | $\Theta(\sqrt{T})$                              |
| partial  | unknown, $\leq K_{\max}$ ,<br>block revelation | $\tilde{O}(nK_{\max}^{4/3}T^{2/3} + K_{\max}B)$ |

Table 2: The landscape. Rows 1–3 are Theorems 5.1, 6.1 and 7.1 of Balcan et al. [2]; row 4 is Theorem 6; row 5 is Theorem 12; row 6 is Proposition 14 with Theorem 6 at  $k = 2$ ; row 7 is Theorem 18. Rows 3 and 5 are incomparable: the construction of row 3 plays  $T$  distinct types, so it needs  $K_{\max} \geq T$  and is consistent with Theorem 6, whereas row 5 uses  $|K| = 1$  but needs partial information and an infinite universe. The  $\Omega(\sqrt{T})$  half of row 6 holds under partial information too, but the matching upper bound there is only the  $O(T^{2/3})$  of row 2; closing that gap is an open problem of Section 10.

name the one it faces. Table 2 places this among the known results.

Between the two lies Problem 15, where an oracle periodically reports the *set* of types met so far, a list of names, nothing about when or how often. Theorem 18 shows this suffices: Algorithm 2 attains  $O(nK_{\max}^{4/3}T^{2/3} \log^{1/3}(nK_{\max}) + K_{\max}B)$ , the partial-information rate of Balcan et al. [2, Theorem 6.1] degraded by  $K_{\max}^{1/3}$  for the unknown catalogue and by an additive  $K_{\max}B$  for the batching, the latter tight by Theorem 22. Corollary 21 makes this a deployment statement: the report needs to arrive only  $\tilde{O}(T^{1/3})$  times over a horizon of length  $T$ , and, by the matching bound there, not fewer.

What Theorem 22 isolates is a *latency* cost: an oracle that speaks only every  $B$  rounds still costs  $\Theta(K_{\max}B)$ , because the defender plays blind between a type’s arrival and the report naming it. This places the additive term alongside the literature on online learning under delayed feedback [10], with the difference that here the delay applies not to the loss but to the *identity* of the opponent, and that Theorem 12 makes the undelayed identity indispensable rather than merely helpful.

Three questions are left open.

**Is the exponent  $2/3$  necessary here?** The exponent in Theorem 18 is inherited from Balcan et al. [2, Theorem 6.1] and untouched by the unknown catalogue, so improving it would improve their result. Whether  $T^{2/3}$  is optimal for partial-information feedback even with  $K$  known is not known; the natural attack replaces their explore-then-exploit estimator by importance-weighted estimates on the exploit rounds, which needs a version of the clean-epochs part of Lemma 17 for a continuously updated estimator.

**Is the factor  $K_{\max}^{1/3}$  necessary?** It comes from Hölder over the epochs, exactly as the  $\sqrt{K_{\max}}$  of Theorem 6 comes from Cauchy–Schwarz. Theorem 22 bounds only the additive term and Proposition 14 only the  $T$ -dependence; a bound  $\Omega(K_{\max}^c T^{2/3})$  matching the multiplicative price is open under both feedback models. The obvious attempt,  $K_{\max}/2$  disjoint copies of the two-type construction of Proposition 14, one per epoch, fails instructively. Per-epoch lower bounds do not add, because the benchmark Equation (3) is a *single* fixed coverage vector: with all intervals disjoint, Lemma 10 says no  $\mathbf{p}$  hits more than one type, so the global benchmark is far weaker than the sum of the per-epoch ones and the global regret correspondingly smaller. A matching lower bound must let the epochs share a good comparator, as the nested family of Theorem 22 does, while still forcing fresh learning in each.

The complementary attack is on the upper bound: avoid the restart and both factors go with it, since without epochs there is neither a Cauchy–Schwarz nor a Hölder step. Algorithms tracking a *growing* expert set [16] do that, and Section 2.2 already notes them as a drop-in for the Polynomial-Weights sub-routine. But replacing the sub-routine is not enough: Steps 4 and 5 of Section 6 also use the epoch structure to pass from the epoch-local optimum to the global  $\mathbf{p}^*$ . Under partial information it is worse still, the estimator of Lemma 24(ii) is defined relative to a fixed catalogue, its spanner and ambient set both changing when  $C$  grows, so estimates from before and after a discovery block are not estimates of comparable quantities.

**Confounded search:  $K_{\max} \geq 2$  on a finite universe.** For  $K_{\max} = 1$  the picture is complete: on a resolution- $M$  universe a miss halves the version space unambiguously, so Remark 13 gives an optimum of  $\Theta(\min\{T, \log_2 M\})$ . For  $K_{\max} \geq 2$  the

defender does not know which opponent produced the observation, so the same search is corrupted. Proposition 14 settles one half, the answer must contain  $\Omega(\sqrt{T})$ , already at  $M = 2$ , and the two upper bounds we can write down,  $O(nK_{\max}\sqrt{T\log(nK_{\max})})$  and  $O(T^{2/3}nM\log^{1/3}(nM))$ , contain no  $\log M$  while the lower bound contains no  $M$  at all. The joint dependence is open.

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## A The random-walk bound of Proposition 14

Let  $S_T = \sum_{t \leq T} \varepsilon_t$  with  $\varepsilon_t = \pm 1$  independent and uniform. We claim  $\mathbb{E}|S_T| \geq \sqrt{\lfloor T/2 \rfloor}$ . Since  $|S_m|$  is a submartingale,  $\mathbb{E}|S_T| \geq \mathbb{E}|S_{2\nu}|$  for  $\nu = \lfloor T/2 \rfloor$ , and the classical identity  $\mathbb{E}|S_{2\nu}| = 2\nu \binom{2\nu}{\nu} 4^{-\nu}$  reduces the claim to

$$\binom{2\nu}{\nu} 4^{-\nu} \geq \frac{1}{2\sqrt{\nu}}. \quad (53)$$

This holds with equality at  $\nu = 1$  and follows by induction: the ratio of consecutive left-hand sides is  $(2\nu+1)/(2\nu+2)$  and of right-hand sides  $\sqrt{\nu/(\nu+1)}$ , and  $(2\nu+1)^2(\nu+1) \geq \nu(2\nu+2)^2$  reduces to  $\nu+1 \geq 0$ . Hence  $\mathbb{E}|S_T| \geq 2\nu/(2\sqrt{\nu}) = \sqrt{\nu}$ .

## B An oracle-efficient implementation

For a catalogue  $C$  and  $\mathbf{p} \in \mathcal{P}$  define the *lift*

$$y_{\alpha,i}(\mathbf{p}) = \mathbb{I}\{b_{\alpha}(\mathbf{p}) = i\}, \quad w_{\alpha,i}(\mathbf{p}) = y_{\alpha,i}(\mathbf{p}) p_i \quad (54)$$

for  $(\alpha, i) \in C \times N$ , and let  $\mathcal{S}(C) = \{(y(\mathbf{p}), w(\mathbf{p})) : \mathbf{p} \in \mathcal{P}\} \subseteq \mathbb{R}^{2|C|n}$ .

**Lemma 24** (both algorithms are linear over the lift). *Assume the types in  $C$  have pairwise distinct best-response maps, so that the set  $\mathcal{W} = \{\mathbb{I}_{(\sigma=i)}\}$  of Balcan et al. [2, Section 6.3] has rank  $|C|$  and its barycentric spanner  $\Lambda$  is invertible. Let  $\omega \in \mathbb{R}^C$  be any vector of weights on the catalogue. Then*

$$\begin{aligned} \sum_{\alpha \in C} \omega_{\alpha} U_d(b_{\alpha}(\mathbf{p}), \mathbf{p}) &= \sum_{\alpha \in C} \sum_{i \in N} \left[ \omega_{\alpha} u_d^u(i) y_{\alpha,i}(\mathbf{p}) \right. \\ &\quad \left. + \omega_{\alpha} (u_d^c(i) - u_d^u(i)) w_{\alpha,i}(\mathbf{p}) \right], \end{aligned} \quad (55)$$

which is a linear function of the lift. Moreover:

1. the loss used by Algorithm 1 at round  $t$  is  $\ell_t(\mathbf{p}) = -U_d(b_{a_t}(\mathbf{p}), \mathbf{p})$ , i.e. Equation (55) with  $\omega = -\mathbb{I}_{\{a_t\}}$ ;
2. the estimator  $\hat{c}_{\tau}$  of Balcan et al. [2, Eq. 2 and Lemma 6.4], used inside Algorithm 2, is Equation (55) with  $\omega = g_{\tau}$ , and the loss it induces is its negation, i.e.  $\omega = -g_{\tau}$ , where  $g_{\tau} = \Lambda^{-1} \hat{p}_{\tau}$ ,  $\Lambda$  is the matrix whose rows are the barycentric spanner  $b_1, \dots, b_{|C|}$  of Balcan et al. [2, Section 6.3] and  $\hat{p}_{\tau} \in \{0, 1\}^{|C|}$  collects that block's exploration outcomes. The vector  $g_{\tau}$  is an unbiased estimator of the per-round type frequencies in the block.

*Proof.* For each  $\alpha$  exactly one of  $y_{\alpha,1}, \dots, y_{\alpha,n}$  equals 1, so  $U_d(b_{\alpha}(\mathbf{p}), \mathbf{p}) = \sum_i y_{\alpha,i}(\mathbf{p}) U_d(i, \mathbf{p})$ . By Equation (1),

$$\begin{aligned} y_{\alpha,i} U_d(i, \mathbf{p}) &= u_d^c(i) y_{\alpha,i} p_i + u_d^u(i) y_{\alpha,i} (1 - p_i) \\ &= u_d^u(i) y_{\alpha,i} + (u_d^c(i) - u_d^u(i)) w_{\alpha,i}, \end{aligned}$$

and summing against  $\omega$  gives Equation (55). Item (i) is the special case  $\omega = -\mathbb{I}_{\{a_t\}}$ .

For (ii), recall that Balcan et al. [2, Eq. 2] sets  $\hat{c}_{\tau}(\mathbf{p}) = \sum_i \sum_j \lambda_j(\mathbb{I}_{(\sigma=i)}) \hat{p}_{\tau}(b_j) U_d(i, \mathbf{p})$ , where  $\sigma$  is the response profile of  $\mathbf{p}$  and  $\lambda(w)$  is the (unique, since  $\Lambda$  is invertible) vector of coefficients with  $\sum_j \lambda_j(w) b_j = w$ , i.e.  $\lambda(w)^{\top} \Lambda = w^{\top}$ . Hence

$\sum_j \lambda_j(w) \hat{p}_{\tau}(b_j) = \lambda(w) \cdot \hat{p}_{\tau} = w^{\top} \Lambda^{-1} \hat{p}_{\tau} = w \cdot g_{\tau}$ . The  $\alpha$ -th coordinate of  $\mathbb{I}_{(\sigma=i)}$  is exactly  $y_{\alpha,i}(\mathbf{p})$ , so

$$\begin{aligned} \hat{c}_{\tau}(\mathbf{p}) &= \sum_i \left( \sum_{\alpha} g_{\tau,\alpha} y_{\alpha,i}(\mathbf{p}) \right) U_d(i, \mathbf{p}) \\ &= \sum_{\alpha} g_{\tau,\alpha} U_d(b_{\alpha}(\mathbf{p}), \mathbf{p}), \end{aligned}$$

which is Equation (55) with  $\omega = g_{\tau}$ . Finally, Balcan et al. [2, Lemma 6.3] gives  $\mathbb{E}[\hat{p}_{\tau}(b_j)] = f_{\tau}(b_j)/|B_{\tau}|$  and  $f_{\tau}(w) = w \cdot m_{\tau}$  with  $m_{\tau}$  the vector of type counts in the block, so  $\mathbb{E}[\hat{p}_{\tau}] = \Lambda m_{\tau}/|B_{\tau}|$  and  $\mathbb{E}[g_{\tau}] = m_{\tau}/|B_{\tau}|$ .  $\square$

Lemma 24 says that both algorithms are instances of online linear optimization over  $\mathcal{S}(C)$ , whose single primitive is

$$\text{ORACLE}(A, \Delta) = \arg \max_{\mathbf{p} \in \mathcal{P}} \sum_{\alpha, i} (A_{\alpha, i} y_{\alpha, i}(\mathbf{p}) + \Delta_{\alpha, i} w_{\alpha, i}(\mathbf{p})). \quad (56)$$

We therefore replace the Polynomial-Weights sub-routine of Proposition 3 by Follow the Perturbed Leader over  $\mathcal{S}(C)$ , which calls Equation (56) once per round, and no times at all on the exploration rounds of Algorithm 2, which play a fixed spanner strategy. This is a change of implementation and not of algorithm: Balcan et al. [2, Section 3.1] explicitly allow any black box satisfying their Proposition 1, and name Follow the Lazy Leader of Kalai and Vempala [11] as an admissible choice. We run it with the perturbation schedule of Kalai and Vempala [11], in its anytime form because the segment lengths are not known in advance. The three problem constants that schedule needs take, on  $\mathcal{S}(C)$ , the values  $\|Y - Y'\|_1 \leq 4|C|$  for the  $\ell_1$ -diameter of the decision set,  $|Y \cdot s| \leq 1$  for the per-round loss, and  $\|s\|_1 \leq 3n$  for the  $\ell_1$ -norm of a one-round loss vector, the last because  $\sum_i (|u_d^u(i)| + |u_d^c(i) - u_d^u(i)|) \leq 3n$ .

Grouping Equation (56) by the response profile  $\sigma : C \rightarrow N$  makes the objective linear on each cell  $\mathcal{P}_{\sigma}(C)$  of Equation (6), so Equation (56) is a maximum of  $n^{|C|}$  linear programs in  $n$  variables, polynomial in  $n$  for a fixed catalogue, and the natural extension to a set of types of the multiple-LP method of Conitzer and Sandholm [5]. In the experiments we instead solve one mixed-integer program with  $n|C|$  binaries in the style of the Bayesian-Stackelberg MILP of Paruchuri et al. [17], using the profile enumeration as a correctness reference. We do not claim polynomial time in  $|C|$ : the Bayesian Stackelberg problem is NP-hard [5]. The claim is polynomial time in  $n$ , which is the parameter that made  $\mathcal{E}(C; \epsilon)$  unusable. Both back-ends solve over cells shrunk by a strictness margin  $\gamma = 10^{-6}$ , so that the returned  $\mathbf{p}$  realizes its profile under any tie-breaking rule; this plays the role of the  $\epsilon$ -approximation in Lemma 1. The rank hypothesis of Lemma 24 is not an extra assumption on the game but a normalisation of the catalogue: two types with the same best-response map on all of  $\mathcal{P}$  are indistinguishable to the defender's payoff, to the benchmark Equation (3) and to the feedback, so Algorithm 2 keeps one representative of each best-response map in  $C$  and discards the duplicates. This can only shrink the catalogue, so  $|C_j| \leq K'$  and every bound of Section 8 is unaffected; without it  $\mathcal{W}$  can have rank below  $|C|$  and  $\Lambda^{-1}$  in Lemma 24(ii) need not exist. The implementation does not de-duplicate but detects the failure: if the sampled pool admits no full-rank spanner it records a rank failure for that catalogue and

skips exploration until the catalogue next changes. Over the runs behind Figure 6 no rank failure occurred, which is what one expects of independently drawn utility vectors. As in Section 9, the experiments take  $\mathcal{P} = \{\mathbf{p} \geq 0 : \sum_i p_i = 1\}$  with one resource; the algorithm and the benchmark  $\mathbf{p}^*$  optimize over the same set, so the regret they define is the one of Equation (4). Every non-asymptotic claim of Sections 7 and 8, together with the oracle and estimator claims of this subsection, is re-checked by `code/verify_theory.py`, which prints a pass or fail for each; the measured quantities of Appendices B.1, C.1 and C.2 and Section 9.4, fitted exponents, envelope constants and timings, are printed by the plotting and benchmarking scripts that produce the corresponding figure or table, and are reproduced by rerunning those.

One further gap between Algorithm 2 and its implementation should be stated. The set  $\mathcal{W} = \{\mathbb{I}_{(\sigma=i)}\}$  of Balcan et al. [2, Section 6.3] has one element per pair (cell, target) and we do not enumerate it; instead we sample coverage vectors, read off the vectors  $\mathbb{I}_{(\sigma=i)}$  they realize together with their witnesses  $(\mathbf{p}_b, i_b)$ , and run the greedy determinant maximization of Awerbuch and Kleinberg [1] on that pool. The consequence is confined to the *range* of the estimator: a spanner of a sub-pool need not give coefficients  $\lambda \in [-1, 1]$  for every  $w \in \mathcal{W}$ , so the bound  $\hat{c}_\tau \in [-n|C|, n|C|]$  of Balcan et al. [2, Lemma 6.4] is not guaranteed by the implementation. Unbiasedness, Lemma 24(ii), which is what the estimator’s correctness rests on, needs only that  $\Lambda$  be invertible, and holds exactly; it is the check reported in the previous paragraph. For the same reason the implementation does not scale the Follow-the-Perturbed-Leader perturbation by that worst-case range: inside Algorithm 2 it uses the schedule  $\epsilon_t = \sqrt{4|C|/(3nt)}$  listed above, which is what the range of the estimates we actually compute warrants, rather than the range  $n|C|$  that Balcan et al. [2, Lemma 6.4] proves in the worst case. This is a tuning choice and not a claim: Theorem 18 rests on Lemma 19 and on their Lemma 6.4, not on the constants used here, and what Figure 6 sets out to test is whether the predicted *exponent* appears in a run. Over the 60 random instances with  $n \in \{3, 4, 5\}$  and  $|C| \in \{1, 2, 3\}$  that `verify_theory.py` runs, the two back-ends and a brute-force maximization over the enumerated  $\mathcal{E}(C; \epsilon)$  agree to within  $1.4 \times 10^{-6}$ , and a longer run of 150 instances does not push that figure up. We also checked Lemma 24(ii) directly: over the  $4 \times 10^4$  replications of a single estimation block at  $n = 6$ ,  $|C| = 3$  that the same script runs, the mean of  $|B_\tau| g_\tau$  reproduced the true type counts of the block to within 0.05 out of 40, comfortably inside the three-standard-error band of 1.1 that Monte-Carlo noise allows; a separate run at  $2 \times 10^5$  replications gave a largest error of 0.09 against a standard error of 0.16, which is what an unbiased estimator should do.

## B.1 Tractability

Table 3 times the three routines. Building  $\mathcal{E}(C; \epsilon)$  already takes 307 seconds at  $n = 5$ ,  $|C| = 4$  and is out of reach beyond that, whereas a single oracle call costs 133 ms at  $n = 10$ , 0.9 s at  $n = 20$  and 1.6 s at  $n = 40$ . Since neither algorithm makes more than one oracle call per round, and Algorithm 2 makes none at all on its exploration rounds, this is what moves the experiments from  $n = 3$  to  $n = 20$ .

| $n$ | $ C $ | $\binom{H}{n-1}$     | build $\mathcal{E}$               | $n^{ C }$ LPs | MILP   |
|-----|-------|----------------------|-----------------------------------|---------------|--------|
| 3   | 4     | $1.1 \times 10^2$    | 6 ms ( $ \mathcal{E}  = 61$ )     | 32 ms         | 16 ms  |
| 5   | 4     | $1.5 \times 10^5$    | 307 s ( $ \mathcal{E}  = 13118$ ) | 230 ms        | 64 ms  |
| 10  | 4     | $7.3 \times 10^{14}$ | infeasible                        | 3.1 s         | 133 ms |
| 20  | 4     | $5.9 \times 10^{37}$ | infeasible                        | 61 s          | 0.88 s |
| 30  | 4     | $1.4 \times 10^{63}$ | infeasible                        | infeasible    | 0.89 s |
| 40  | 4     | $1.2 \times 10^{90}$ | infeasible                        | infeasible    | 1.56 s |

Table 3: Cost of one step, with  $H = H(n, |C|) = n + |C| \binom{n}{2}$ . Column 3 is the number of  $(n-1)$ -subsets that the exhaustive construction of  $\mathcal{E}(C; \epsilon)$  must solve. Columns 5 and 6 are the two back-ends for Equation (56); wherever both ran they returned the same optimum to  $2 \times 10^{-6}$ . Median of 10 calls on an otherwise idle machine; `code/bench_oracle.py`. The MILP is not monotone in  $n$  because it is an exact branch-and-bound solve, not a fixed amount of work.

## C Experiments at $n = 10$ and $n = 20$

### C.1 Full information at $n = 10$ and $n = 20$

Figure 5 reruns the two experiments of Section 9 with the oracle implementation, at  $n = 10$  and  $n = 20$  instead of  $n = 3$ , with  $K_{\max} = 4$  and  $T = 1500$ . Panels (a) and (b) reproduce Figure 2 under the staggered-uniform adversary: the per-round regret spikes shortly after each of the four phase boundaries, a new type becomes available at a boundary and is first drawn a few rounds later, which is the discovery round, and then decays inside the epoch that follows, exactly the structure that Steps 1–6 of the proof of Theorem 6 describe. Panels (c) and (d) reproduce Figure 3 under the uniform-random adversary: the cumulative regret stays below the  $C\sqrt{t}$  envelope of Theorem 6 throughout, with  $C = 1.27$  at  $n = 10$  and  $C = 1.71$  at  $n = 20$ , and it is 8.0 and 11.6 times smaller than the uniform baseline.

We keep the two adversaries separate for the same reason Section 9 does, and it is worth spelling out. The staggered-uniform adversary is non-stationary by design, the type mixture changes at every phase boundary, so the best *fixed* strategy in hindsight of Equation (3) is a weak comparator, and an algorithm that tracks the current phase can and does beat it. At  $n = 10$  the cumulative regret against  $\mathbf{p}^*$  came out negative on one of the four seeds ( $-35.3$ ), and the four-seed mean was  $-2.6$ . That says nothing against Theorem 6, which is an upper bound, but it makes a cumulative-regret plot on that adversary uninformative. The uniform-random adversary is stationary, so  $\mathbf{p}^*$  is the right comparator, and panels (c) and (d) use it. The benchmark  $\mathbf{p}^*$  is itself computed with one oracle call, so even *evaluating* the regret at  $n = 20$  was out of reach before Appendix B.

### C.2 Block revelation at $n = 10$ and $n = 20$

Figure 6 runs Algorithm 2 under genuine partial-information feedback: the simulator passes the algorithm the attacked target only, and at the end of each revelation block calls it with the *set* of types seen so far and nothing else. The set is handed over as a list for convenience, but the algorithm reads it only through membership tests, so neither the order in which the types first attacked nor any frequency information

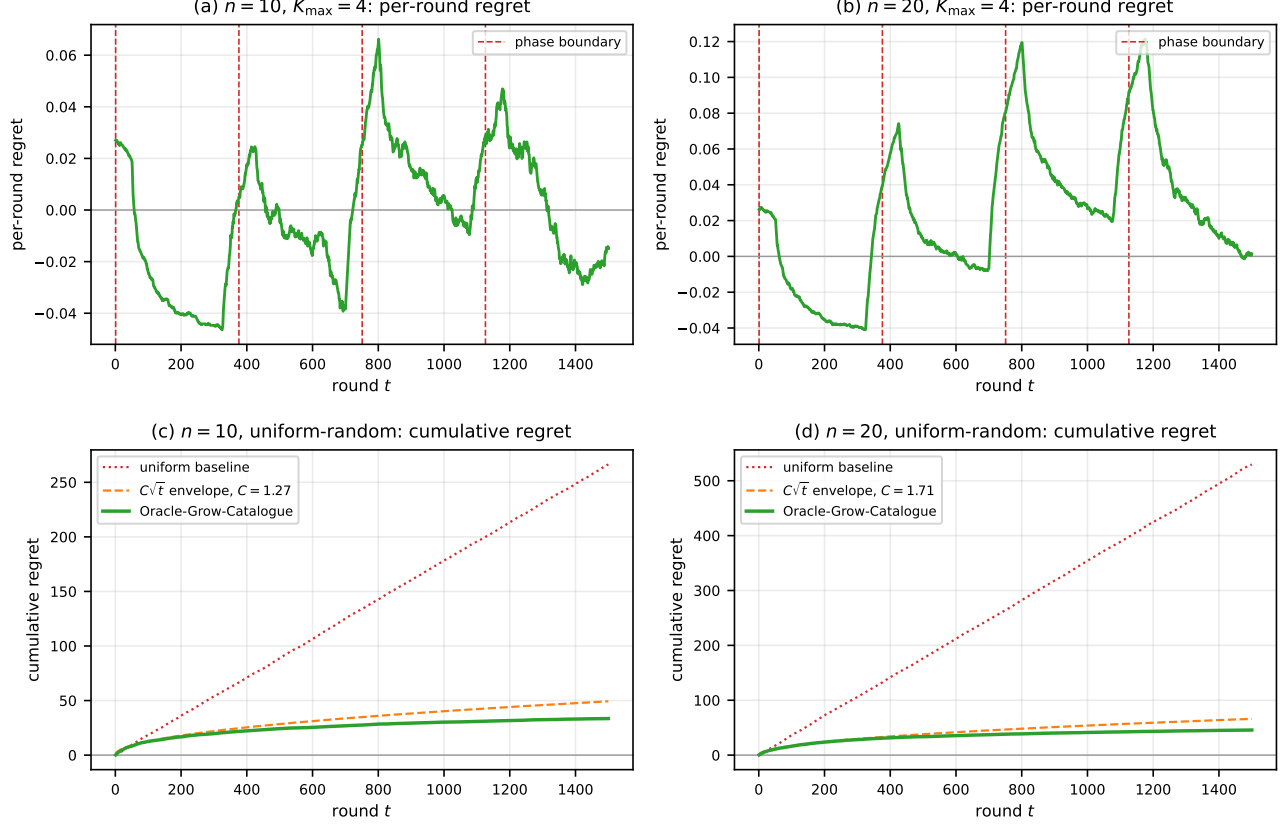


Figure 5: Algorithm 1 with the oracle implementation of Appendix B,  $K_{\max} = 4$ ,  $T = 1500$ , mean over 4 seeds. (a), (b) per-round regret under the staggered-uniform adversary, rolling window 101; red dashed lines mark the phase boundaries, at which a new type becomes available. (c), (d) cumulative regret under the uniform-random adversary, against the  $C\sqrt{t}$  envelope of Theorem 6 and the uniform baseline.

can leak through it. Panels (a) and (b) show cumulative regret against the  $Ct^{2/3}$  envelope of Theorem 18 under the staggered-uniform adversary, which is the interesting one here because it spreads the discovery blocks across the horizon and so produces the  $\Theta(K_{\max})$  epochs that Step 8 of the proof pays for; the envelope constants are  $C = 1.72$  at  $n = 10$  and  $C = 1.25$  at  $n = 20$ .

Panel (c) sweeps the revelation length  $B$  over six doublings at fixed  $T$ . The measured growth is linear in  $B$ , as Theorems 18 and 22 predict, a least-squares fit over  $B \in [4, 256]$  has slope 0.47, but that slope is far below the worst-case  $2K' = 8$ , and the panel draws both. The gap is expected rather than a discrepancy: the worst case is attained only by the nested adversary of Theorem 22, which Figure 4(c) exhibits separately, and on a random instance most of the rounds of a discovery block are not in fact wasted, because a strategy tuned to the types already in the catalogue is usually not terrible against a new one.

Panel (d) compares the two feedback models as  $T$  grows, on the stationary uniform-random adversary for the reason given in Appendix C.1. Over  $T \in \{250, 500, 1000, 2000\}$  the block-revelation regret grows with fitted exponent 0.664, which is  $2/3$  to within the resolution of the experiment and is exactly the exponent of Theorem 18; the full-information regret grows with

exponent 0.317, comfortably inside the  $1/2$  of Theorem 6. The asymmetry is not an accident of the instances. Algorithm 2 pays a floor of  $|C|$  exploration rounds in every estimation block whatever the sequence does, so its  $T^{2/3}$  is visible even on an easy sequence, whereas Algorithm 1 has no such floor and, as already observed in Section 9, runs well inside its worst case against a stochastic adversary.

One implementation choice is worth stating, because it is where asymptotics and simulation disagree. Balcan et al. [2, Algorithm 1] uses  $Z = n(T^2 \log(n|C|))^{1/3}$  estimation blocks, i.e. blocks of  $T/Z = T^{1/3}/(n \log^{1/3}(n|C|))$  rounds, which at  $n = 10$ ,  $|C| = 4$  and  $T = 2000$  is less than one round, fewer, certainly, than the  $|C| = 4$  exploration rounds a block must hold, since the tuning of Balcan et al. [2, Theorem 6.1] is asymptotic. We therefore use blocks of  $\max\{|C| + 1, \lceil T^{1/3} \rceil\}$  rounds, which has the same order in  $T$ . At  $T = 2000$  and  $K_{\max} = 4$  that is 13 rounds, of which 4 are spent exploring, so roughly 31% of all rounds are exploration; this is the main reason the measured regret in panels (a) and (b) is an order of magnitude above the full-information regret of Figure 5 at the same horizon, and it is a constant-factor effect, not a change of exponent.

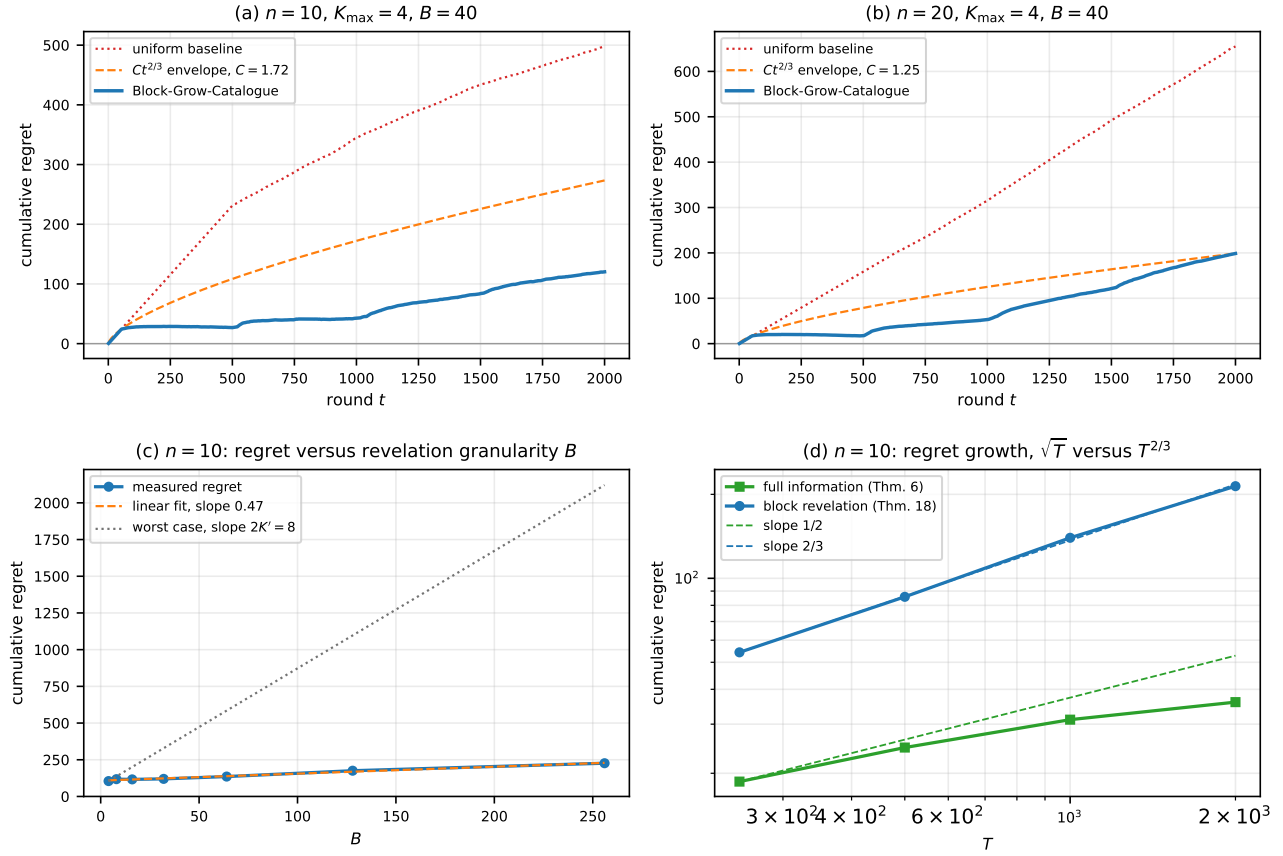


Figure 6: Algorithm 2 under partial-information feedback with block revelation,  $K_{\max} = 4$ ,  $T = 2000$ , mean over 4 seeds. (a)–(c) staggered-uniform adversary; (d) uniform-random adversary. (a), (b) cumulative regret against the  $Ct^{2/3}$  envelope of Theorem 18. (c) final regret as a function of the revelation length  $B$ , against a least-squares fit and against the worst-case slope  $2K'$  of Theorem 18. (d) full information (Theorem 6) against block revelation (Theorem 18) as  $T$  grows; fitted exponents 0.317 and 0.664.